

O R A C L E A P E X T U T O R I A L S

Oracle APEX

Interview Q&A

2026 Edition — Updated for APEX 26.1

234 Questions**3 Levels** Entry · Mid · Expert

AI Agents · APEX 26.1

APEX • SQL • PL/SQL • JavaScript • CSS • AI Agents •
Security • Performance

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HOW TO USE THIS GUIDE

Each question shows a level badge — **ENTRY**, **INTERMEDIATE**, or **EXPERIENCED**. The teal ANSWER block gives the core explanation. Code examples appear in the shaded box below.

Cover the answer, attempt it yourself, then compare. That is the fastest way to prepare.

APEX Architecture

Q1

ENTRY
LEVEL

APEX ARCHITECTURE

What is Oracle APEX and what does it run on?

ANSWER

Oracle APEX (Application Express) is a free, low-code web development platform built on Oracle Database. You build full web applications directly in your browser — no separate IDE needed. It runs entirely inside Oracle Database and uses ORDS (Oracle REST Data Services) as the web listener.

EXAMPLE

```
// Developer logs into apex.oracle.com → Workspace
// Builds Leave Management app via drag-and-drop
// No Java, Node.js, or separate IDE required
// ORDS handles all HTTP communication
```

Q2

ENTRY
LEVEL

APEX ARCHITECTURE

What is the 3-tier architecture of Oracle APEX?

ANSWER

1. Client Tier — the user's browser renders HTML, CSS, and JS.
2. Middle Tier — ORDS handles HTTP requests and routes them to the database.
3. Database Tier — Oracle Database stores all APEX metadata, application code, session state, and data.

EXAMPLE

Browser → ORDS → APEX Engine (inside DB) → Tables → Response

Q3

ENTRY
LEVEL

APEX ARCHITECTURE

What is ORDS and why is it needed in APEX?

ANSWER

ORDS is the middle-tier web listener. It receives HTTP/HTTPS requests from the browser, passes them to the APEX engine inside Oracle Database, and returns the HTML response. Without ORDS, APEX cannot be accessed over the web. ORDS also auto-generates REST endpoints when you enable a table for REST in SQL Workshop.

EXAMPLE

```
GET /ords/vendix/customers/
POST /ords/vendix/customers/
GET /ords/vendix/customers/:id
-- Auto-generated when you REST-enable a table in SQL Workshop
```

Q4

ENTRY
LEVEL

APEX ARCHITECTURE

What is a Workspace in Oracle APEX?

ANSWER

A Workspace is a shared development area inside an APEX instance. Multiple developers can share one workspace. Each workspace is mapped to one or more Oracle database schemas. It contains applications, files, SQL scripts, and user accounts.

EXAMPLE

```
-- Example:  
Workspace: VENDIX_WS  
Mapped schema: VENDIX_SCHEMA  
Developer Ravi builds POS apps inside this workspace
```

Q5

ENTRY
LEVEL

APEX ARCHITECTURE

What is the difference between a Workspace Administrator and a Developer in APEX?

ANSWER

Workspace Administrator: manages users, schemas, storage, and workspace settings.

Developer: can only build and edit applications. Cannot create new users or change workspace-level settings.

APEX Core Components

Q6

ENTRY
LEVEL

APEX CORE COMPONENTS

What is Application Builder used for?

ANSWER

Application Builder is the main interface inside APEX for creating and editing applications. From here you create pages, regions, items, dynamic actions, validations, processes, and navigation — all from the browser. No external tool needed.

Q7

ENTRY
LEVEL

APEX CORE COMPONENTS

What is SQL Workshop in APEX?

ANSWER

SQL Workshop is a browser-based database tool inside APEX. It contains:

- Object Browser — view tables, triggers, sequences
- SQL Commands — run SQL/PL/SQL interactively
- SQL Scripts — save and run scripts
- Data Workshop — import/export CSV and Excel

Pages & Regions

Q8

ENTRY
LEVEL

PAGES & REGIONS

What is a Page in APEX and what components does it have?

ANSWER

A Page is one screen of an APEX application — the basic building block. Components: Regions, Page Items, Buttons, Dynamic Actions, Processes, Validations, Computations, and Branches.

EXAMPLE

Page 1 = Login page

Page 2 = Dashboard

Page 3 = Customer List (Interactive Report)

Q9

ENTRY
LEVEL

PAGES & REGIONS

What is a Region in Oracle APEX? Name 5 region types.

ANSWER

A Region is a container inside a page that holds content.

Region types: (1) Form, (2) Interactive Report, (3) Interactive Grid, (4) Classic Report, (5) Chart, (6) Cards, (7) Calendar, (8) Static Content, (9) PL/SQL Dynamic Content.

Q10

ENTRY
LEVEL

PAGES & REGIONS

What is a Page Item in APEX? Name at least 6 item types.**A N S W E R**

A Page Item is a UI field on a page that stores a value in session state.

Types: (1) Text Field, (2) Select List, (3) Date Picker, (4) Checkbox, (5) Radio Group, (6) Textarea, (7) Hidden, (8) Password, (9) File Browse, (10) Rich Text Editor.

E X A M P L E

-- P3_CUSTOMER_NAME is a Text Field on page 3

-- Referenced in SQL/PL/SQL as:

```
SELECT * FROM CUSTOMERS WHERE NAME = :P3_CUSTOMER_NAME
```

Session State

Q11

ENTRY
LEVEL

SESSION STATE

What is Session State in APEX and how does it work?**A N S W E R**

Session State is APEX's mechanism to store and retrieve page item values across requests. When a page is submitted, APEX saves all item values in session state in the database. Access values using bind variable syntax :P1_ITEM or substitution &P1_ITEM.

E X A M P L E

-- User enters name on Page 1 → P1_NAME stored in session state

-- On Page 2, filter using that stored value:

```
SELECT * FROM ORDERS WHERE CUSTOMER_NAME = :P1_NAME
```

Q12

ENTRY
LEVEL

SESSION STATE

How do you pass a value from one APEX page to another?**A N S W E R**

Three ways:

1. URL parameter — append &P2_ITEM=value in the branch or link.
2. Application Item — store in a global item visible across all pages.
3. APEX Collection — for temporary multi-row data.

E X A M P L E

-- URL with parameter:

```
f?p=100:2:&SESSION:::NO::P2_EMP_ID:101
```

Pages & Regions

Q13**ENTRY
LEVEL****PAGES & REGIONS**

What is a Page Process in APEX? When does it fire?

ANSWER

A Page Process is server-side logic (PL/SQL, DML, or built-in) that runs during page rendering or page submission.

- Rendering processes fire when the page loads.
- Processing processes fire when the page is submitted.

EXAMPLE

```
-- Insert Customer process on form submit:
BEGIN
  INSERT INTO CUSTOMERS(NAME, EMAIL)
  VALUES(:P3_NAME, :P3_EMAIL);
END;
```

Q14**ENTRY
LEVEL****PAGES & REGIONS**

What is a Validation in APEX? What types exist?

ANSWER

A Validation is an edit check that runs before page processing. If it fails, APEX shows the page again with an error — page processes do NOT run.

Types: (1) Not Null, (2) Item Is Numeric, (3) SQL Expression, (4) PL/SQL Function, (5) Regular Expression.

EXAMPLE

```
-- Validate P3_EMAIL using Regular Expression:
-- Pattern: ^[a-zA-Z0-9._%+-]+@[a-zA-Z0-9.-]+\.[a-zA-Z]{2,}$
```

Q15**ENTRY
LEVEL****PAGES & REGIONS**

What is a Branch in APEX? When is it used?

ANSWER

A Branch controls where the user goes after page submission. It can redirect to a page, URL, or procedure. You set conditions to decide which branch fires.

EXAMPLE

```
-- After saving customer form → branch to Page 3
```

- If validation failed → no branch fires
- Same page reloads with error messages

Dynamic Actions

Q16**ENTRY
LEVEL****DYNAMIC ACTIONS**

What is a Dynamic Action in APEX? Give a real example.

ANSWER

A Dynamic Action is a declarative way to add client-side interactivity without writing JavaScript manually. You define: WHEN (event) → CONDITION (optional) → TRUE ACTION.

EXAMPLE

- No JS code needed:
- WHEN: P3_COUNTRY changes
- TRUE ACTION: Refresh P3_STATE select list
- APEX generates all client-side JS automatically

LOV & Items

Q17**ENTRY
LEVEL****LOV & ITEMS**

What is a LOV in APEX? Difference between Static and Dynamic LOV?

ANSWER

A List of Values (LOV) provides options for select lists, radio groups, and checkboxes.

- Static LOV: hardcoded values (ACTIVE;Active, INACTIVE;Inactive).
- Dynamic LOV: values come from a SQL query and update based on data.

EXAMPLE

```
-- Dynamic LOV SQL:  
SELECT DEPT_NAME d, DEPT_ID r  
FROM DEPARTMENTS  
ORDER BY DEPT_NAME
```

Reports

Q18**ENTRY
LEVEL****REPORTS**

What is the difference between a Classic Report and an Interactive Report?

ANSWER

Classic Report: Simple, static tabular report. No user customization. Best for print-ready output.

Interactive Report (IR): Users can filter, sort, group, highlight, pivot, chart, and save custom views. Best for data exploration and analysis.

Q19**ENTRY
LEVEL****REPORTS**

What is an Interactive Grid? How is it different from an Interactive Report?

ANSWER

Interactive Grid combines IR flexibility with editable spreadsheet-style functionality. Users can edit cells inline, add rows, delete rows — directly on the page.

- IR: read-only for end users.
 - IG: allows DML (Insert/Update/Delete) from the UI.
- APEX 26.1 added clipboard support (cut/copy/paste/drag-drop) to IGs.

EXAMPLE

- Use IR for: Sales reports, dashboards (read-only)
- Use IG for: Editable grids, budget tables, pricing editors

Pages & Regions

Q20**ENTRY
LEVEL****PAGES & REGIONS**

What is a Computation in APEX and when does it run?

ANSWER

A Computation sets a value for a page item automatically. It runs either Before Header (on page load) or After Submit. You can use SQL, PL/SQL, or a static value.

EXAMPLE

- Compute P1_TODAY = SYSDATE before header
- Result: date field always shows today's date on page load

Shared Components

Q21**ENTRY
LEVEL**

SHARED COMPONENTS

What are Shared Components in APEX? Name 5 examples.

A N S W E R

Shared Components are reusable elements defined once and used across multiple pages.

Examples: (1) LOVs, (2) Navigation Menus, (3) Authorization Schemes, (4) Authentication Schemes, (5) Application Items, (6) Web Credentials, (7) Generative AI Services.

Q22**ENTRY
LEVEL**

SHARED COMPONENTS

What is a Substitution String in APEX? Give 4 real examples.

A N S W E R

Substitution Strings are APEX variables replaced with actual values at runtime. They use & prefix and . suffix in HTML/JS.

E X A M P L E

&APP_USER. → logged-in username (e.g., RAVI)

&APP_ID. → application ID (e.g., 100)

&APP_SESSION. → current session ID

&P1_CUST_NAME. → value of page item P1_CUST_NAME

-- In HTML region:

Welcome, &APP_USER.! → renders as: Welcome, RAVI!

Q23**ENTRY
LEVEL**

SHARED COMPONENTS

What is the difference between an Application Item and a Page Item?

A N S W E R

Page Item: scoped to one specific page. P3_NAME only exists on page 3.

Application Item: global — accessible from any page, process, or SQL without a page prefix. Used for user roles, settings, or tenant IDs needed across the whole app.

E X A M P L E

-- Application Item:

G_USER_ROLE = 'ADMIN' -- set at login, checked on every page

-- Page Item:

P3_CUSTOMER_NAME -- only exists on page 3

Navigation

Q24**ENTRY
LEVEL****NAVIGATION**

What is a Navigation Menu in APEX and how is it configured?

ANSWER

A Navigation Menu is the sidebar or top navigation in Universal Theme apps. Defined as a List in Shared Components. Each entry has: Name, Page Target, Icon, and a Condition to show/hide based on user role.

Security

Q25**ENTRY
LEVEL****SECURITY**

What is an Authorization Scheme in APEX?

ANSWER

An Authorization Scheme controls access to a page, region, item, or button. It returns TRUE (access allowed) or FALSE (access denied). The same scheme can be applied to multiple components.

EXAMPLE

```
-- IS_ADMIN authorization scheme:  
RETURN APEX_UTIL.IS_IN_GROUP('ADMIN');  
-- Apply to admin-only pages, regions, and buttons
```

Q26**ENTRY
LEVEL****SECURITY**

What is an Authentication Scheme in APEX?

ANSWER

An Authentication Scheme controls HOW users log in — it verifies identity.

Types: (1) Application Express Accounts, (2) Custom PL/SQL, (3) LDAP/Active Directory, (4) Social Sign-In (Google, IDCS), (5) No Authentication (public app).

Q27

ENTRY
LEVEL

SECURITY

What is the difference between Authentication and Authorization in APEX?

ANSWER

Authentication = WHO ARE YOU? Verifies identity at login. Runs once per session.

Authorization = WHAT CAN YOU ACCESS? Checks permissions per page/component. Runs on every page request.

EXAMPLE

- Auth flow:
- 1. User logs in → Authentication verifies identity
- 2. Inside app → Authorization checks: ADMIN menu visible?
- 3. Regular user: Admin menu hidden via Authorization Scheme

Theme & Templates

Q28

ENTRY
LEVEL

THEME & TEMPLATES

What is a Template in APEX? How does it differ from a Theme?

ANSWER

Theme: A collection of templates defining overall look and feel (colours, fonts, layout).

Template: A specific HTML/CSS/JS structure for one component type (Page, Region, Button, Item).

A Theme contains many Templates.

Q29

ENTRY
LEVEL

THEME & TEMPLATES

What is Universal Theme in APEX?

ANSWER

Universal Theme (Theme 42) is the default, fully responsive, accessible theme shipped with APEX since version 5. It supports mobile-first design, dark mode, Oracle JET components, and customisation with Theme Roller — no CSS writing needed.

APEX Core Components

Q30

ENTRY
LEVEL

APEX CORE COMPONENTS

What is the Page Designer in APEX and what are its 3 panels?

ANSWER

Page Designer is the visual editor for building APEX pages. It has three panels:

1. Left panel — component tree (Regions, Items, Buttons)
2. Centre panel — page canvas with visual layout
3. Right panel — property editor for the selected component

Reports

Q31

ENTRY
LEVEL

REPORTS

What is the Link Column in an Interactive Report used for?

ANSWER

A Link Column adds a clickable link to each row in an Interactive Report — used to navigate to a detail or edit form for that row. Configured with a target page and primary key URL parameter.

EXAMPLE

```
f?p=&APP_ID.:5:&SESSION.:NO::P5_EMP_ID:#EMP_ID#  
-- Navigates to Page 5 Edit form with EMP_ID set from clicked row
```

APEX Core Components

Q32

ENTRY
LEVEL

APEX CORE COMPONENTS

What is a Card Region in APEX?

ANSWER

A Card Region displays data in a visual card layout. Each row becomes a card showing an icon, title, subtitle, body text, and badge. Defined using a SQL query mapped to visual attributes.

EXAMPLE

```
SELECT PROD_NAME, PRICE, STOCK_QTY, CATEGORY_ICON  
FROM PRODUCTS  
-- Map: Title=PROD_NAME, Badge=PRICE, Body=STOCK_QTY, Icon=CATEGORY_ICON
```

Q33**ENTRY
LEVEL****APEX CORE COMPONENTS**

What is a Chart Region in APEX? What chart types are supported?

ANSWER

A Chart Region renders data visually using Oracle JET Charts. The source is a SQL query returning label and value columns.

Chart types: Bar, Line, Area, Pie, Donut, Scatter, Bubble, Funnel, Gauge, Pyramid, Combo.

EXAMPLE

```
SELECT DEPT_NAME, COUNT(*) AS EMP_COUNT
FROM EMPLOYEES
GROUP BY DEPT_NAME
```

Pages & Regions

Q34**ENTRY
LEVEL****PAGES & REGIONS**

What is the difference between a Form region and a Form page?

ANSWER

Form Page: APEX wizard creates a full page with Form region + Submit/Cancel/Delete buttons pre-wired to DML processes. Best for a dedicated Create/Edit page.

Form Region: A form embedded inside any page, coexisting with other regions. You wire DML processes manually.

APEX Core Components

Q35**ENTRY
LEVEL****APEX CORE COMPONENTS**

What are Template Options in APEX? Give 3 real examples.

ANSWER

Template Options are predefined CSS modifier classes in Universal Theme that change appearance without writing custom CSS. Applied in the Template Options dialog in region or button properties.

EXAMPLE

```
-- Region:
Style = 'Accent 1'  → coloured header bar
Header = 'Hidden'   → hides region title
-- Button:
Style = 'Success'   → green button
```

SQL Fundamentals

Q36

ENTRY
LEVEL

SQL FUNDAMENTALS

Write a SQL query to fetch top 5 records from a table.

ANSWER

Use FETCH FIRST (Oracle 12c+) — the modern standard. ROWNUM is the legacy approach.

EXAMPLE

-- Modern (Oracle 12c+):

```
SELECT * FROM EMPLOYEES FETCH FIRST 5 ROWS ONLY;
```

-- Legacy:

```
SELECT * FROM EMPLOYEES WHERE ROWNUM <= 5;
```

Q37

ENTRY
LEVEL

SQL FUNDAMENTALS

Write a query to find the second highest salary from EMP table.

ANSWER

Use a subquery excluding the maximum. DENSE_RANK() approach handles ties correctly.

EXAMPLE

-- Option 1: Subquery

```
SELECT MAX(SAL) FROM EMP  
WHERE SAL < (SELECT MAX(SAL) FROM EMP);
```

-- Option 2: DENSE_RANK (handles ties)

```
SELECT SAL FROM (  
  SELECT SAL, DENSE_RANK() OVER (ORDER BY SAL DESC) RNK  
  FROM EMP  
) WHERE RNK = 2;
```

Q38

ENTRY
LEVEL

SQL FUNDAMENTALS

What is the difference between WHERE and HAVING clauses?

ANSWER

WHERE: filters rows BEFORE grouping. Cannot use aggregate functions.

HAVING: filters groups AFTER GROUP BY. Can use aggregate functions like SUM, COUNT.

SQL Joins

Q39**ENTRY
LEVEL****SQL JOINS**

What is the difference between INNER JOIN, LEFT JOIN, RIGHT JOIN, and FULL OUTER JOIN?

A N S W E R

INNER JOIN: Returns only rows matching in BOTH tables.

LEFT JOIN: Returns ALL rows from left + matching from right (NULLs where no match).

RIGHT JOIN: Returns ALL rows from right + matching from left.

FULL OUTER JOIN: Returns ALL rows from BOTH tables with NULLs where no match.

E X A M P L E

-- LEFT JOIN: shows ALL employees, even with no department:

```
SELECT E.ENAME, D.DNAME  
FROM EMP E  
LEFT JOIN DEPT D ON E.DEPTNO = D.DEPTNO;
```

SQL Fundamentals

Q40**ENTRY
LEVEL****SQL FUNDAMENTALS**

What is a subquery? Write a real example.

A N S W E R

A subquery is a SELECT statement nested inside another SQL statement. Used in WHERE, FROM, or SELECT clauses.

E X A M P L E

-- Employees earning more than average salary:

```
SELECT ENAME, SAL  
FROM EMP  
WHERE SAL > (SELECT AVG(SAL) FROM EMP);
```

Q41

ENTRY
LEVEL

SQL FUNDAMENTALS

What is the difference between UNION and UNION ALL?**A N S W E R**

UNION: Combines results and removes duplicate rows. Slower (deduplication sort).

UNION ALL: Combines results and KEEPS all duplicates. Faster — no deduplication.

E X A M P L E

```
SELECT ENAME FROM EMP WHERE DEPTNO = 10
UNION ALL
SELECT ENAME FROM EMP WHERE DEPTNO = 20;
```

Q42

ENTRY
LEVEL

SQL FUNDAMENTALS

What is GROUP BY in SQL? When is it used?**A N S W E R**

GROUP BY groups rows sharing the same value in specified columns. Always used with aggregate functions (SUM, COUNT, AVG, MAX, MIN) to calculate a value per group.

E X A M P L E

```
SELECT DEPTNO, AVG(SAL) AS AVG_SALARY, COUNT(*) AS HEADCOUNT
FROM EMP
GROUP BY DEPTNO
ORDER BY DEPTNO;
```

SQL DDL & DML

Q43

ENTRY
LEVEL

SQL DDL & DML

What is the difference between DELETE, TRUNCATE, and DROP?**A N S W E R**

DELETE: Removes specific rows. DML — can be rolled back. Fires triggers.

TRUNCATE: Removes ALL rows. DDL — cannot be rolled back. Does NOT fire triggers. Very fast.

DROP: Removes the entire table structure AND data permanently.

E X A M P L E

```
DELETE FROM EMP WHERE DEPTNO = 10; -- rollback possible
TRUNCATE TABLE EMP;               -- fast, no rollback
DROP TABLE EMP;                   -- removes table permanently
```


SQL Constraints

Q44**ENTRY
LEVEL****SQL CONSTRAINTS**

What is a PRIMARY KEY vs a UNIQUE KEY?

A N S W E R

PRIMARY KEY: Uniquely identifies each row. Cannot be NULL. Only ONE per table. Auto-creates an index.

UNIQUE KEY: Values must be unique but CAN be NULL (one NULL per column in Oracle). Multiple per table allowed.

Q45**ENTRY
LEVEL****SQL CONSTRAINTS**

What is a FOREIGN KEY? Write an example.

A N S W E R

A FOREIGN KEY is a column referencing the PRIMARY KEY of another table. Enforces referential integrity — you cannot insert a value that does not exist in the parent table.

E X A M P L E

```
CREATE TABLE ORDERS (  
  ORDER_ID NUMBER PRIMARY KEY,  
  CUSTOMER_ID NUMBER REFERENCES CUSTOMERS(CUST_ID),  
  ORDER_DATE DATE DEFAULT SYSDATE  
);
```

SQL Functions

Q46**ENTRY
LEVEL****SQL FUNCTIONS**

What are aggregate functions in Oracle? Name 5 with examples.

A N S W E R

Aggregate functions calculate a result over a set of rows:

(1) COUNT(*), (2) SUM(SAL), (3) AVG(SAL), (4) MAX(SAL), (5) MIN(SAL).

E X A M P L E

```
SELECT COUNT(*), SUM(SAL), AVG(SAL), MAX(SAL), MIN(SAL)  
FROM EMP  
WHERE DEPTNO = 10;
```

Q47

ENTRY
LEVEL

SQL FUNCTIONS

What is the NVL function? Write an example.

ANSWER

NVL(expr1, expr2) replaces NULL with a default. Returns expr2 if expr1 is NULL, otherwise returns expr1.

EXAMPLE

```
-- Show 0 instead of NULL for commission:
SELECT ENAME, NVL(COMM, 0) AS COMMISSION
FROM EMP;
-- Employees with no commission show 0, not NULL
```

Q48

ENTRY
LEVEL

SQL FUNCTIONS

What is the difference between COUNT(*) and COUNT(column_name)?

ANSWER

COUNT(*): Counts ALL rows including those with NULL values.

COUNT(column_name): Counts only rows where that column is NOT NULL.

EXAMPLE

```
-- EMP has 14 rows; only 4 have a COMM value:
SELECT COUNT(*), COUNT(COMM) FROM EMP;
-- Result: 14    4
```

SQL Objects

Q49

ENTRY
LEVEL

SQL OBJECTS

What is a VIEW in Oracle? Why would you use one?

ANSWER

A VIEW is a stored SQL query behaving like a virtual table. It does NOT store data — it executes the query when accessed.

Uses: (1) Simplify complex queries, (2) Security layer hiding sensitive columns, (3) Stable interface for reporting.

EXAMPLE

```
CREATE OR REPLACE VIEW V_EMP_DEPT AS
SELECT E.ENAME, E.SAL, D.DNAME
FROM EMP E JOIN DEPT D ON E.DEPTNO = D.DEPTNO;

SELECT * FROM V_EMP_DEPT WHERE SAL > 5000;
```

Q50

ENTRY
LEVEL

SQL OBJECTS

What is a SEQUENCE in Oracle and how is it used in APEX?

ANSWER

A SEQUENCE generates unique numbers automatically. In APEX, sequences generate primary key values when inserting new records.

EXAMPLE

```
CREATE SEQUENCE EMP_SEQ START WITH 1 INCREMENT BY 1;
```

```
INSERT INTO EMP (EMP_ID, ENAME)  
VALUES (EMP_SEQ.NEXTVAL, 'Ravi');
```

SQL Data Types

Q51

ENTRY
LEVEL

SQL DATA TYPES

What is the difference between CHAR, VARCHAR2, and NVARCHAR2 in Oracle?

ANSWER

CHAR(n): Fixed-length, always stores exactly n characters (space-padded). Best for fixed codes.

VARCHAR2(n): Variable-length, stores only actual characters. Most commonly used.

NVARCHAR2(n): Variable-length Unicode string for international characters.

EXAMPLE

CHAR(2) → stores 'Y ' (padded) -- YES/NO flags

VARCHAR2(100) → stores 'Ravi' (4 chars) -- most common

NVARCHAR2(50) → stores Hindi/Arabic text -- Unicode

SQL Fundamentals

Q52**ENTRY
LEVEL****SQL FUNDAMENTALS**

What is the DUAL table in Oracle?

A N S W E R

DUAL is a special single-row, single-column dummy table. Used when you need to SELECT a value that does not come from any real table.

E X A M P L E

```
SELECT SYSDATE FROM DUAL;      -- get current date
SELECT UPPER('hello') FROM DUAL; -- test a function
SELECT 2 + 2 FROM DUAL;        -- arithmetic
SELECT EMP_SEQ.NEXTVAL FROM DUAL; -- next sequence value
```

SQL Functions

Q53**ENTRY
LEVEL****SQL FUNCTIONS**

What is TO_DATE and TO_CHAR in Oracle SQL?

A N S W E R

TO_DATE: Converts a string to a DATE data type.

TO_CHAR: Converts a DATE or NUMBER to a formatted string.

E X A M P L E

```
-- String to Date:
SELECT TO_DATE('25-JUN-2026','DD-MON-YYYY') FROM DUAL;

-- Date to formatted String:
SELECT TO_CHAR(SYSDATE,'DD/MM/YYYY HH24:MI:SS') FROM DUAL;
-- Output: 25/06/2026 14:30:00
```

Q54

ENTRY
LEVEL

SQL FUNCTIONS

What is the difference between SYSDATE and SYSTIMESTAMP?**A N S W E R**

SYSDATE: Returns current database date and time without fractional seconds. Data type = DATE.

SYSTIMESTAMP: Returns current date and time WITH fractional seconds and time zone. Data type = TIMESTAMP WITH TIME ZONE.

E X A M P L E

```
SELECT SYSDATE, SYSTIMESTAMP FROM DUAL;  
-- SYSDATE:    25-JUN-26  
-- SYSTIMESTAMP: 25-JUN-26 02.30.45.123456 PM +05:30
```

Q55

ENTRY
LEVEL

SQL FUNCTIONS

What is the DECODE function in Oracle?**A N S W E R**

DECODE(expr, search1, result1, search2, result2, ..., default) compares expr against each search value and returns the corresponding result. Oracle-specific. For modern code, prefer CASE WHEN.

E X A M P L E

```
SELECT DECODE(DEPTNO,  
  10, 'Accounting',  
  20, 'Research',  
  30, 'Sales',  
  'Other') AS DEPT_NAME  
FROM EMP;
```

PL/SQL Basics

Q56

ENTRY
LEVEL

PL/SQL BASICS

What is the difference between a PL/SQL Procedure and a Function?**A N S W E R**

Procedure: Performs an action. Does NOT return a value via RETURN. Can have IN, OUT, IN OUT parameters. Cannot be used directly in SQL.

Function: Returns exactly one value via RETURN. Can be used inside SQL expressions.

Q57

ENTRY
LEVEL

PL/SQL BASICS

What is a PL/SQL block structure? Explain DECLARE / BEGIN / EXCEPTION / END.**A N S W E R**

A PL/SQL block has 4 sections:

1. DECLARE — optional. Declare variables, cursors, types.
2. BEGIN — mandatory. The actual executable code.
3. EXCEPTION — optional. Handle runtime errors.
4. END — mandatory. Marks end of block.

E X A M P L E

```
DECLARE
  v_name VARCHAR2(50);
BEGIN
  SELECT ENAME INTO v_name FROM EMP WHERE EMPNO = 7369;
  DBMS_OUTPUT.PUT_LINE('Name: ' || v_name);
EXCEPTION
  WHEN NO_DATA_FOUND THEN
    DBMS_OUTPUT.PUT_LINE('Employee not found');
END;
```

Q58

ENTRY
LEVEL

PL/SQL BASICS

What is a Cursor in PL/SQL? Difference between Implicit and Explicit?**A N S W E R**

A Cursor is a pointer to a result set in memory.

Implicit Cursor: Created automatically by Oracle for SELECT INTO, INSERT, UPDATE, DELETE. Attributes: SQL%FOUND, SQL%ROWCOUNT.

Explicit Cursor: Declared by the developer for multi-row queries. Full control over OPEN, FETCH, CLOSE.

E X A M P L E

```
DECLARE
  CURSOR c_emp IS SELECT ENAME, SAL FROM EMP;
  v_rec c_emp%ROWTYPE;
BEGIN
  OPEN c_emp;
  LOOP
    FETCH c_emp INTO v_rec;
    EXIT WHEN c_emp%NOTFOUND;
    DBMS_OUTPUT.PUT_LINE(v_rec.ENAME || ':' || v_rec.SAL);
  END LOOP;
  CLOSE c_emp;
END;
```

Q59

ENTRY
LEVEL

PL/SQL BASICS

What is an Exception in PL/SQL? What is WHEN OTHERS?

ANSWER

An Exception is a runtime error in PL/SQL. Oracle provides predefined exceptions (NO_DATA_FOUND, TOO_MANY_ROWS, ZERO_DIVIDE) and you can define custom ones.

WHEN OTHERS: Catches any unhandled exception. Always put it LAST in the EXCEPTION block.

EXAMPLE

```
EXCEPTION
WHEN NO_DATA_FOUND THEN
  DBMS_OUTPUT.PUT_LINE('No rows found');
WHEN TOO_MANY_ROWS THEN
  DBMS_OUTPUT.PUT_LINE('More than one row returned');
WHEN OTHERS THEN
  DBMS_OUTPUT.PUT_LINE('Error: ' || SQLERRM);
```

Q60

ENTRY
LEVEL

PL/SQL BASICS

What is %TYPE and %ROWTYPE in PL/SQL?

ANSWER

%TYPE: Declares a variable with the same data type as a specific table column.

%ROWTYPE: Declares a variable that holds an entire row of a table.

EXAMPLE

```
-- %TYPE — same datatype as EMP.SAL:
v_sal EMP.SAL%TYPE;

-- %ROWTYPE — holds a full row of EMP:
v_emp EMP%ROWTYPE;

SELECT * INTO v_emp FROM EMP WHERE EMPNO = 7369;
DBMS_OUTPUT.PUT_LINE(v_emp.ENAME || ' earns ' || v_emp.SAL);
```

Q61

ENTRY
LEVEL

PL/SQL BASICS

What is a Package in PL/SQL? What are its advantages?

ANSWER

A Package groups related procedures, functions, variables, and cursors.

Advantages: (1) Better organization, (2) Session-persistent variables, (3) Overloading, (4) Performance (loaded once per session), (5) Encapsulation.

EXAMPLE

```
CREATE OR REPLACE PACKAGE pkg_emp AS
```

```
PROCEDURE add_emp(p_name VARCHAR2);  
FUNCTION get_sal(p_id NUMBER) RETURN NUMBER;  
END pkg_emp;
```

```
-- Call:  
v_sal := pkg_emp.get_sal(7369);
```

Q62**ENTRY
LEVEL****PL/SQL BASICS**

What is a Trigger in Oracle? When would you use one?

ANSWER

A Trigger fires AUTOMATICALLY when a DML or DDL event occurs on a table.

Types: BEFORE, AFTER, INSTEAD OF (for views).

Use cases: Audit logging, auto-populating CREATED_BY/CREATED_DATE, enforcing business rules.

EXAMPLE

```
CREATE OR REPLACE TRIGGER TRG_EMP_AUDIT  
BEFORE UPDATE ON EMP  
FOR EACH ROW  
BEGIN  
  INSERT INTO EMP_AUDIT(EMP_ID, OLD_SAL, NEW_SAL, CHANGED_BY)  
  VALUES(:OLD.EMPNO, :OLD.SAL, :NEW.SAL, USER);  
END;
```

Q63**ENTRY
LEVEL****PL/SQL BASICS**

What is DBMS_OUTPUT.PUT_LINE used for?

ANSWER

DBMS_OUTPUT.PUT_LINE writes messages to a buffer displayed in SQL Developer or SQL Workshop (when server output is ON). Used for debugging PL/SQL during development. NOT available in APEX runtime — use APEX_DEBUG.MESSAGE instead.

EXAMPLE

```
-- In SQL Workshop/SQL Developer:  
BEGIN  
  DBMS_OUTPUT.PUT_LINE('Processing: ' || :P3_EMP_ID);  
END;  
  
-- In APEX runtime — use this instead:  
APEX_DEBUG.MESSAGE('Processing: %s', :P3_EMP_ID);
```


Q64

ENTRY
LEVEL

PL/SQL BASICS

How do you handle NULL values in PL/SQL?

ANSWER

Any comparison with NULL returns NULL (not TRUE or FALSE). Always use IS NULL / IS NOT NULL. Use NVL() to replace NULL with a default.

EXAMPLE

```
-- Check for NULL:
IF v_comm IS NULL THEN v_comm := 0; END IF;

-- Shorter with NVL:
v_comm := NVL(v_comm, 0);

-- NVL2: different result for NULL vs NOT NULL:
v_label := NVL2(v_comm, 'Has Commission', 'No Commission');
```

Q65

ENTRY
LEVEL

PL/SQL BASICS

Write a simple FOR loop in PL/SQL.

ANSWER

A PL/SQL FOR loop automatically manages the loop counter. You do not need to declare it. Add REVERSE keyword for a countdown loop.

EXAMPLE

```
BEGIN
  FOR i IN 1..5 LOOP
    DBMS_OUTPUT.PUT_LINE('Iteration: ' || i);
  END LOOP;
END;

-- Reverse loop:
BEGIN
  FOR i IN REVERSE 1..5 LOOP
    DBMS_OUTPUT.PUT_LINE('Count down: ' || i);
  END LOOP;
END;
```

HTML & CSS

Q66

ENTRY
LEVEL

HTML & CSS

What is the difference between id and class in HTML/CSS?

ANSWER

id: Unique identifier. Only ONE element per page can have a specific id. Used in CSS as #myId. In APEX, item names become element IDs (P3_NAME → id='P3_NAME').

class: Can be shared by multiple elements. Used in CSS as .myClass.

EXAMPLE

```
<!-- id: unique, one per page -->
<div id='dashboard-header'>Header</div>

<!-- class: reusable on many elements -->
<div class='kpi-card'>Card 1</div>
<div class='kpi-card'>Card 2</div>
```

Q67

ENTRY
LEVEL

HTML & CSS

What is the CSS box model? Explain margin, border, padding, content.

ANSWER

Every HTML element is a rectangular box with 4 layers from inside out:

1. Content — actual text/image area
2. Padding — space between content and border
3. Border — outline around the element
4. Margin — space outside the border (between elements)

EXAMPLE

```
.kpi-card {
  width: 200px; /* content */
  padding: 10px; /* inside space */
  border: 2px solid #0d9488; /* outline */
  margin: 15px; /* outside space */
}
/* Total rendered width = content + padding + border + margin */
```

JavaScript

Q68

ENTRY
LEVEL

JAVASCRIPT

What does `document.getElementById()` do in JavaScript?

ANSWER

`document.getElementById('id')` returns a reference to the HTML element with that specific id. In APEX, prefer `apex.item()` for page items — it handles all item types correctly.

EXAMPLE

```
// Standard JS (limited):  
var el = document.getElementById('P3_CUST_NAME');  
console.log(el.value);
```

```
// APEX API (preferred — works for all item types):  
var val = apex.item('P3_CUST_NAME').getValue();
```

Q69

ENTRY
LEVEL

JAVASCRIPT

How do you show or hide an HTML element using JavaScript?

ANSWER

Use `display` or `visibility` CSS properties. In APEX, use Dynamic Actions (Show/Hide action) instead of manual JS for page items and regions — it handles all item types correctly.

EXAMPLE

```
// Standard JS:  
document.getElementById('myDiv').style.display = 'none'; // hide  
document.getElementById('myDiv').style.display = 'block'; // show
```

```
// APEX preferred — via Dynamic Action or:  
apex.item('P3_CONTAINER').hide();  
apex.item('P3_CONTAINER').show();
```

Q70

ENTRY
LEVEL

JAVASCRIPT

What is the difference between `var`, `let`, and `const` in JavaScript?

ANSWER

`var`: Function-scoped. Can be redeclared. Hoisted. Old style — avoid in new code.

`let`: Block-scoped. Cannot be redeclared in same scope. Use for mutable variables.

`const`: Block-scoped. Cannot be reassigned after declaration. Use for values that should not change.

E X A M P L E

```
const APP_VERSION = '26.1'; // never changes
let errorCount = 0;        // will be updated

if (condition) {
  let blockVar = 'only here'; // block scoped
} // blockVar doesn't exist outside the if block
```

HTML & CSS**Q71****ENTRY
LEVEL****HTML & CSS****What is the difference between inline, internal, and external CSS?****A N S W E R**

Inline CSS: Written in element's style attribute. Highest priority. Hard to maintain.

Internal CSS: In <style> tags inside <head>. Good for one page.

External CSS: Separate .css file. Best for reuse across pages.

In APEX: use Page CSS for page-specific styles; upload .css to Static Application Files for app-wide CSS.

E X A M P L E

```
<!-- Inline (avoid for maintainability) -->
<div style='color:red;'>Text</div>

/* APEX page-level CSS (Page Properties → CSS → Inline) */
.my-region { border-radius: 8px; }

/* APEX app-level: Static Application Files */
:root { --vdx-primary: #0d9488; }
```

Q72**ENTRY
LEVEL****HTML & CSS****What is a CSS selector? Name 4 types.****A N S W E R**

CSS selectors target HTML elements to apply styles.

- (1) Element: p, div, h1
- (2) Class: .card, .button-primary
- (3) ID: #dashboard-header
- (4) Attribute: input[type='text']
- (5) Pseudo-class: a:hover, button:focus

E X A M P L E

```
/* Element */
p { font-family: Arial; }

/* Class */
.kpi-card { border-radius: 8px; }

/* ID */
#P3_EMAIL { border: 2px solid #0d9488; }

/* Pseudo-class */
.btn-primary:hover { opacity: 0.9; }
```

Q73**ENTRY
LEVEL****HTML & CSS**

What is the difference between `display:none` and `visibility:hidden` in CSS?

ANSWER

`display:none` — Element is completely removed from the layout. No space reserved. Page reflows.

`visibility:hidden` — Element is invisible BUT still occupies space in the layout. No reflow.

In APEX Dynamic Actions, Show/Hide uses `display:none` by default.

EXAMPLE

```
/* display:none — removed from flow, no space kept */
.hidden-region { display: none; }

/* visibility:hidden — invisible but keeps space */
.invisible-keeper { visibility: hidden; }

/* APEX Dynamic Action handles this automatically */
/* Use DAs instead of manually toggling CSS classes */
```

IR vs IG

Q74

INTERMEDIATE

IR VS IG

What is the difference between IR and IG in a real project context?

ANSWER

Interactive Report (IR): Read-only. Users filter, sort, group, chart, export, save views. Best for data analysis.

Interactive Grid (IG): Editable spreadsheet-style. Users add, edit, delete rows inline. APEX 26.1 added clipboard support (cut/copy/paste/drag-drop). Best for data entry.

EXAMPLE

- Use IR for: Sales analysis, dashboards (read-only)
- Use IG for: Order entry, budget allocation, pricing editor
- IG requires: single-table SQL + Primary Key column defined

Forms & Master-Detail

Q75

INTERMEDIATE

FORMS & MASTER-DETAIL

How do you implement a master-detail form in APEX?

ANSWER

1. Create a Form Region for the master (e.g., Customer).
2. Add an Interactive Grid for detail records (e.g., Orders).
3. Link the IG: WHERE CUST_ID = :P5_CUST_ID in the IG SQL.
4. Set IG to refresh when the master item changes using a Dynamic Action.

EXAMPLE

```
-- Detail IG SQL linked to master P5_CUST_ID:
SELECT ORDER_ID, ORDER_DATE, AMOUNT, STATUS
FROM ORDERS
WHERE CUST_ID = :P5_CUST_ID
ORDER BY ORDER_DATE DESC
```

AJAX Callbacks

Q76**INTERMEDIATE****AJAX CALLBACKS**

What is an AJAX Callback in APEX and how do you create and call one?

ANSWER

An AJAX Callback is a server-side PL/SQL process that runs without a full page reload. Created as a Process with point 'Ajax Callback'. Called from JavaScript using `apex.server.process()`. Used for dynamic data fetch and background updates.

EXAMPLE

```
-- PL/SQL AJAX Callback (name: GET_PRICE):
DECLARE v_price NUMBER; BEGIN
SELECT PRICE INTO v_price FROM PRODUCTS
WHERE ID = :P3_PROD_ID;
APEX_JSON.OPEN_OBJECT;
APEX_JSON.WRITE('price', v_price);
APEX_JSON.CLOSE_OBJECT;
END;

// JavaScript call:
apex.server.process('GET_PRICE',
{ pagemItems: '#P3_PROD_ID' },
{ success: r => apex.item('P3_PRICE').setValue(r.price) });
```

Regions & Refresh

Q77**INTERMEDIATE****REGIONS & REFRESH**

How do you refresh a region without a full page reload in APEX?

ANSWER

Use a Dynamic Action: Event → True Action = Refresh → Selection Type = Region → select your region. This re-runs the region SQL and re-renders only that region.

EXAMPLE

```
-- Declarative DA approach:
Event: Button Click [SEARCH_BTN]
True Action: Refresh
Affected Region: Customer Report

// JavaScript approach:
apex.region('customer_report_static_id').refresh();
```

Dynamic Actions

Q78**INTERMEDIATE****DYNAMIC ACTIONS**

How do you conditionally show/hide a page item using Dynamic Actions (no JS)?

ANSWER

Create a Dynamic Action:

- WHEN: Change event on P3_CATEGORY.
- True Action: Show — affected items: P3_SUBCATEGORY.
- False Action: Hide — affected items: P3_SUBCATEGORY.

APEX generates all client-side JS automatically.

EXAMPLE

- Zero JavaScript needed:
- DA: When P3_CATEGORY = 'ELECTRONICS'
- True: Show P3_BRAND, Show P3_WATTS
- False: Hide P3_BRAND, Hide P3_WATTS

Page Processing

Q79**INTERMEDIATE****PAGE PROCESSING**

What is the Page Rendering vs Page Processing order in APEX? Why does sequence matter?

ANSWER

Page Rendering (page loads): Before Header → After Header → Regions → Footer.

Page Processing (page submits): Computations → Validations → Processes → Branches.

Sequence number controls order within each point. A Computation setting a value must have a lower sequence number than the Validation checking that value.

EXAMPLE

- Wrong: Validation (seq 10) fires BEFORE Computation (seq 20)
- P3_STATUS is still NULL when validation runs!

- Correct:
- Computation: set P3_STATUS → Sequence 5
- Validation: check P3_STATUS → Sequence 10

Security

Q80**INTERMEDIATE****SECURITY**

How do you implement row-level security in an Interactive Report?

ANSWER

Add a WHERE clause to the IR SQL that filters rows based on the logged-in user (:APP_USER). This ensures each user sees only their own data. For admin users, show all rows using a role check.

EXAMPLE

```
SELECT TICKET_ID, SUBJECT, STATUS, PRIORITY
FROM TICKETS
WHERE ASSIGNED_TO = :APP_USER
   OR :APP_USER IN (
       SELECT USERNAME FROM ADMIN_USERS
     )
ORDER BY CREATED_ON DESC
```

Faceted Search

Q81**INTERMEDIATE****FACETED SEARCH**

What is a Faceted Search in APEX? When would you use it instead of an IR?

ANSWER

Faceted Search provides a filter panel where users click grouped filter options (like e-commerce — Category, Price Range, Brand). Results update instantly. Unlike IR toolbar filters, Faceted Search shows filter panels on the side.

Best for: product catalogues, knowledge bases, searchable directories.

EXAMPLE

```
-- Setup:
-- 1. Create Search Region (source = SQL query)
-- 2. Add Facets: Category (Checkbox), Price (Range)
-- 3. Results region refreshes automatically on facet click
-- No SQL WHERE clause changes needed — APEX handles it
```

Dynamic Actions

Q82**INTERMEDIATE****DYNAMIC ACTIONS**

How do you call a PL/SQL procedure from a Dynamic Action?

A N S W E R

In the Dynamic Action True Action = Execute Server-side Code (PL/SQL). Paste PL/SQL in the code editor. Specify Items to Submit (values sent to server) and Items to Return (values updated after execution).

E X A M P L E

```
-- Items to Submit: P3_CUST_ID, P3_STATUS
BEGIN
  pkg_customers.update_status(
    p_cust_id => :P3_CUST_ID,
    p_status => :P3_STATUS
  );
END;
-- Items to Return: P3_UPDATED_ON
```

Page Processing

Q83**INTERMEDIATE****PAGE PROCESSING**

What is APEX_APPLICATION.G_PRINT_SUCCESS_MESSAGE used for?

A N S W E R

Set this package variable in a PL/SQL process to display a custom green success notification at the top of the page after the process completes.

E X A M P L E

```
BEGIN
  INSERT INTO CUSTOMERS(NAME, EMAIL)
  VALUES(:P3_NAME, :P3_EMAIL);

  APEX_APPLICATION.G_PRINT_SUCCESS_MESSAGE :=
    'Customer ' || :P3_NAME || ' added successfully!';
END;
```

Validations

Q84**INTERMEDIATE****VALIDATIONS**

How do you implement a custom PL/SQL validation in APEX?

ANSWER

Create a Validation: Type = PL/SQL Function (returning error text). Return NULL if valid, or a message string if invalid. APEX shows this inline next to the item.

EXAMPLE

```
DECLARE v_count NUMBER;
BEGIN
  SELECT COUNT(*) INTO v_count
  FROM CUSTOMERS
  WHERE EMAIL = :P3_EMAIL
  AND CUST_ID != NVL(:P3_CUST_ID, 0);
  IF v_count > 0 THEN
    RETURN 'This email is already registered.';
  END IF;
  RETURN NULL; -- NULL means validation passed
END;
```

Reports

Q85**INTERMEDIATE****REPORTS**

How do you create a drill-down report in APEX?

ANSWER

In an IR or Classic Report, configure the Link Column. Set the target page and pass the primary key as a URL parameter. The linked page uses that parameter to display the detail data.

EXAMPLE

```
-- Link Column Target URL:
f?p=&APP_ID.:5:&SESSION.:NO::P5_EMP_ID:#EMP_ID#
-- Page 5 Edit form loads that employee using P5_EMP_ID
```

Security**Q86****INTERMEDIATE****SECURITY****What is Session State Protection (SSP) in APEX and why is it important?****A N S W E R**

SSP generates a cryptographic checksum for URL parameters containing sensitive values. If the URL is tampered with, the checksum fails and APEX throws an error. Critical for preventing users from accessing other users' data by changing IDs in the URL.

E X A M P L E

```
-- Enable per item:  
-- Item → Security → Session State Protection  
-- = 'Restricted - may not be set from browser'  
  
-- URL attack example:  
-- Attacker changes: f?p=100:5:SESSION::NO::P5_CUST_ID:999  
-- APEX detects tampered checksum → blocks request
```

Q87**INTERMEDIATE****SECURITY****How do you implement an authorization scheme based on user roles?****A N S W E R**

1. Create a USER_ROLES table (USERNAME, ROLE_NAME).
2. Create an Authorization Scheme: Type = PL/SQL Function returning Boolean.
3. Check if current user has the required role.
4. Apply the scheme to pages, regions, or buttons.

E X A M P L E

```
-- Authorization Scheme: IS_ADMIN  
DECLARE v_count NUMBER;  
BEGIN  
  SELECT COUNT(*) INTO v_count  
  FROM USER_ROLES  
  WHERE USERNAME = :APP_USER  
    AND ROLE_NAME = 'ADMIN';  
  RETURN v_count > 0;  
END;
```

Q88**INTERMEDIATE****SECURITY**

What are the types of Authentication Schemes available in APEX?

A N S W E R

- (1) Application Express Accounts — managed inside APEX.
- (2) Custom Authentication — PL/SQL validates against your own USERS table.
- (3) LDAP/Active Directory — enterprise directory.
- (4) Oracle IDCS/Social Sign-In — Google, Microsoft OAuth.
- (5) HTTP Header Variable — SSO via proxy.
- (6) No Authentication — public application.

REST Integration

Q89**INTERMEDIATE****REST INTEGRATION**

How do you consume an external REST API in APEX?

A N S W E R

Option 1 (No code): Create a REST Data Source in Shared Components. Use directly as a region source.

Option 2 (PL/SQL): Use APEX_WEB_SERVICE.MAKE_REST_REQUEST(). Full control over request/response.

Option 3: Use a Web Credential for authenticated APIs.

E X A M P L E

```
DECLARE l_response CLOB;
BEGIN
  l_response := APEX_WEB_SERVICE.MAKE_REST_REQUEST(
    p_url      => 'https://api.example.com/products',
    p_http_method => 'GET'
  );
  APEX_JSON.PARSE(l_response);
  :P3_RESULT := APEX_JSON.GET_VARCHAR2('name');
END;
```

Q90**INTERMEDIATE****REST INTEGRATION**

What is a Web Credential in APEX? When is it used?

A N S W E R

A Web Credential is encrypted storage for API keys, passwords, and tokens. The secret is stored encrypted in the APEX repository — NEVER visible in exports or code. Used for: OpenAI API key, OpenRouter API key, any REST API requiring authentication.

E X A M P L E

```
-- Web Credential:
Authentication Type: HTTP Header
Credential Name: Authorization
Secret: Bearer sk-your-api-key ← encrypted
Static ID: AI_POWERED_CRED
-- Never hardcode the key in PL/SQL or page processes
```

File Upload**Q91****INTERMEDIATE****FILE UPLOAD****How do you implement file upload in APEX and store it in the database?****A N S W E R**

1. Add File Browse item (P3_FILE) to the page.
2. On Submit, APEX stores the file in APEX_APPLICATION_FILES (temporary).
3. In a page process, copy from APEX_APPLICATION_FILES to your table.
4. Delete from APEX_APPLICATION_FILES after copying.

E X A M P L E

```
INSERT INTO MY_DOCUMENTS(FILE_NAME, FILE_BLOB, MIME_TYPE, UPLOADED_ON)
SELECT FILENAME, BLOB_CONTENT, MIME_TYPE, SYSDATE
FROM APEX_APPLICATION_FILES
WHERE NAME = :P3_FILE;
```

```
-- Clean up temp storage:
DELETE FROM APEX_APPLICATION_FILES WHERE NAME = :P3_FILE;
```

Email**Q92****INTERMEDIATE****E M A I L****What is APEX_MAIL? How do you send email from APEX?****A N S W E R**

APEX_MAIL is the APEX package for sending emails from PL/SQL. It queues the email and APEX sends it via the configured SMTP server. Use PUSH_QUEUE to send immediately.

E X A M P L E

```
BEGIN
APEX_MAIL.SEND(
  p_to    => 'customer@example.com',
```

```
p_from    => 'noreply@myapp.com',
p_subj    => 'Your Order Confirmation',
p_body    => 'Your order has been placed.',
p_body_html => '<b>Order placed!</b>'
);
APEX_MAIL.PUSH_QUEUE;
END;
```

Deployment

Q93**INTERMEDIATE****DEPLOYMENT**

How do you export and import an APEX application?

ANSWER

Export: Application Builder → Export → Export Application → Download .sql file.

Import: Application Builder → Import → Upload .sql file → Install.

APEX 26.1 also supports APEXlang (.apx) format — human-readable and Git-friendly.

EXAMPLE

```
-- SQLcl export in APEXlang format (26.1):
apex export -applicationid 100 -format apexlang

-- Creates per-page files:
-- app100/pages/page_001.apx
-- app100/shared_components.apx
```

Theme & Templates

Q94**INTERMEDIATE****THEME & TEMPLATES**

What is Theme Roller in APEX? What can you customize with it?

ANSWER

Theme Roller is a visual CSS customization tool inside APEX (Shared Components → Themes → Theme Roller). Customize: colours, fonts, border radius, button styles, spacing — all without writing CSS. Generates a custom.css file stored in the application.

EXAMPLE

```
:root {
  --ut-palette-primary: #0D9488; /* primary color */
  --ut-border-radius: 8px;      /* card rounding */
  --ut-button-padding-y: 10px;  /* button height */
}
```

```
}  
-- Theme Roller generates these CSS variables automatically
```

Q95**INTERMEDIATE****THEME & TEMPLATES**

How do you add custom CSS to an APEX application without breaking the theme?

A N S W E R

Best practices:

1. Page-level CSS: Page Properties → CSS → Inline.
2. App-level CSS: Upload .css to Static Application Files, reference in Theme.
3. Use APEX CSS variables (--ut-*, --a-*) instead of hardcoded hex values.
4. Never edit Universal Theme files directly.

E X A M P L E

```
:root { --brand-teal: #0d9488; }  
  
.my-kpi-card {  
  background: var(--brand-teal);  
  border-radius: var(--ut-border-radius);  
}  
  
/* Avoid hardcoded hex — breaks when theme changes */
```

Debugging

Q96**INTERMEDIATE****DEBUGGING**

What is APEX_DEBUG? How do you use it to troubleshoot a page?

A N S W E R

APEX_DEBUG instruments PL/SQL code with messages visible in APEX debug output. Enable debug: click bug icon in builder or append &p_debug=YES to URL. View messages: Session → View Debug.

E X A M P L E

```
BEGIN  
  APEX_DEBUG.MESSAGE('Start: processing customer %s', :P3_CUST_ID);  
  
  UPDATE CUSTOMERS SET STATUS = :P3_STATUS  
  WHERE CUST_ID = :P3_CUST_ID;  
  
  APEX_DEBUG.MESSAGE('Done: updated %s rows', SQL%ROWCOUNT);  
END;
```


APEX Collections

Q97

INTERMEDIATE

APEX COLLECTIONS

What is an APEX Collection? When would you use one?

ANSWER

An APEX Collection is a temporary in-session data structure (C001-C050 VARCHAR2, N001-N005 NUMBER, CLOB001 CLOB). Data is lost when session ends. Use cases: multi-step wizards, shopping cart, temporary editable datasets before final commit.

EXAMPLE

```
APEX_COLLECTION.CREATE_OR_TRUNCATE_COLLECTION('CART');
APEX_COLLECTION.ADD_MEMBER(
  p_collection_name => 'CART',
  p_c001 => :P3_PROD_ID,
  p_n001 => :P3_QTY
);
```

```
SELECT c001, n001 FROM APEX_COLLECTIONS
WHERE COLLECTION_NAME = 'CART';
```

Q98

INTERMEDIATE

APEX COLLECTIONS

When would you choose an APEX Collection over a Global Temporary Table?

ANSWER

Use APEX Collection when: data is user-specific per session, data is temporary (wizard step data), simplicity is preferred (no DDL).

Use Global Temporary Table (GTT) when: complex queries needed (joins, indexes), data exceeds collection column limits, performance on large datasets is critical.

EXAMPLE

```
-- Collection (simple, session-specific, no DDL):
APEX_COLLECTION.ADD_MEMBER('WIZARD_DATA', p_c001 => :P3_NAME);
```

```
-- GTT (complex queries, better performance):
CREATE GLOBAL TEMPORARY TABLE GTT_WIZARD_DATA (
  NAME VARCHAR2(100), STEP NUMBER
) ON COMMIT PRESERVE ROWS;
```

Wizards**Q99****INTERMEDIATE****WIZARDS****How do you implement a multi-step wizard in APEX?****A N S W E R**

1. Create multiple pages (Step 1, 2, 3...).
2. Use an APEX Collection to store data from each step.
3. On the last step, commit collection data to actual database tables.
4. Use a Wizard Region for visual step indicators.
5. Add Back/Next branches between pages.

E X A M P L E

```
-- Save to collection before advancing to Step 3:
APEX_COLLECTION.UPDATE_MEMBER_ATTRIBUTE(
  p_collection_name => 'SIGNUP',
  p_seq            => 1,
  p_attr_number    => 2,
  p_attr_value     => :P2_COMPANY
);
-- Branch to Page 3
```

REST Integration**Q100****INTERMEDIATE****REST INTEGRATION****What is a REST Data Source vs calling REST via PL/SQL?****A N S W E R**

REST Data Source: Declarative, no-code. Define URL, method, authentication in Shared Components. Use directly as a region source.

REST via PL/SQL (APEX_WEB_SERVICE): Programmatic. Full control over request/response. Needed inside page processes, AJAX callbacks, or triggers.

E X A M P L E

```
-- REST Data Source (declarative — use as region source):
-- Shared Components → REST Data Sources → Create

-- APEX_WEB_SERVICE (programmatic — in a Process):
l_resp := APEX_WEB_SERVICE.MAKE_REST_REQUEST(
  p_url => 'https://api.example.com/products',
  p_http_method => 'GET');
```

APEX Utilities

Q101**INTERMEDIATE****APEX UTILITIES**

What is APEX_UTIL? Give 5 real function examples.

ANSWER

APEX_UTIL is the APEX utility package with many helper functions for session state, user groups, and navigation.

EXAMPLE

```
APEX_UTIL.GET_SESSION_STATE('P3_NAME') -- get item
APEX_UTIL.SET_SESSION_STATE('STATUS','A') -- set item
APEX_UTIL.IS_IN_GROUP('ADMIN')          -- check role
APEX_UTIL.CURRENT_USER_IN_ROLES('ADMIN') -- multi-role
APEX_UTIL.REDIRECT_URL('f?p=100:2:...') -- redirect
```

Dynamic Filtering

Q102**INTERMEDIATE****DYNAMIC FILTERING**

How do you implement search with multiple filters without reloading the page?

ANSWER

1. Create filter items (select lists, text fields).
2. Create a Dynamic Action on each filter item (Change event) → True Action: Refresh the report region.
3. In the report SQL, reference filter items using bind variables with IS NULL fallback.

EXAMPLE

```
SELECT PROD_ID, PROD_NAME, CATEGORY, PRICE
FROM PRODUCTS
WHERE (:P3_CATEGORY IS NULL OR CATEGORY_ID = :P3_CATEGORY)
AND (:P3_MIN_PRICE IS NULL OR PRICE >= :P3_MIN_PRICE)
AND (:P3_SEARCH IS NULL
    OR UPPER(PROD_NAME) LIKE '%' || UPPER(:P3_SEARCH) || '%')
```

Modal Dialogs

Q103

INTERMEDIATE

MODAL DIALOGS

What is the difference between a Modal Dialog page and a regular APEX page?**A N S W E R**

Regular Page: Opens as full browser navigation. User leaves current page.

Modal Dialog Page: Opens as an overlay on top of the current page. Parent page remains visible but inactive. On close/save, the parent refreshes without navigation. Best for forms, confirmations, lookups.

E X A M P L E

```
-- Set page to Modal Dialog:
-- Page → User Interface → Page Mode = Modal Dialog

-- Open from parent page:
-- Button Action = Open a Page in a Dialog
-- Target Page = 10 (your modal page)
```

Q104

INTERMEDIATE

MODAL DIALOGS

How do you pass values TO and FROM a Modal Dialog in APEX?**A N S W E R**

TO modal: Set values using URL parameters in the Open Dialog DA or link.

FROM modal: In the modal's close process, use APEX_UTIL.SET_SESSION_STATE. Parent DA triggered by 'Dialog Closed' event reads the returned values.

E X A M P L E

```
-- In Modal close process:
APEX_UTIL.SET_SESSION_STATE('P3_SELECTED_ID', :P10_PROD_ID);

-- Parent DA: Event = Dialog Closed
-- True Action 1: Refresh Products Grid
-- True Action 2: Set P3_PRICE from P3_SELECTED_PRICE
```

LOV & Cascading

Q105 INTERMEDIATE

LOV & CASCADING

How do you implement cascading LOVs in APEX?

ANSWER

1. Create parent select list (P3_COUNTRY).
 2. Create child select list (P3_STATE) with Dynamic LOV SQL referencing the parent.
 3. In P3_STATE → Cascading LOV Parent Item(s) = P3_COUNTRY.
- APEX automatically refreshes the child when parent changes.

EXAMPLE

```
-- Child LOV SQL:
SELECT STATE_NAME, STATE_ID
FROM STATES
WHERE COUNTRY_ID = :P3_COUNTRY
ORDER BY STATE_NAME
```

UX Helpers

Q106 INTERMEDIATE

UX HELPERS

How do you implement a confirmation dialog before deleting a record?

ANSWER

Simplest: In button properties, set the Confirmation attribute with the message text. APEX renders a browser confirmation dialog automatically when the button is clicked.

EXAMPLE

```
-- Button Properties:
Action:    Submit Page
Confirmation: Are you sure you want to delete this record?
           This action cannot be undone.

-- Alternative: Dynamic Action → Confirm action
-- TRUE: Submit page | FALSE: Cancel
```

IR Advanced

Q107

INTERMEDIATE

IR ADVANCED

What is IR Saved Reports? How do public vs private saved reports differ?**A N S W E R**

Saved Reports let users save customized IR views (filters, sort, columns) with a name.

Private: visible only to the user who created them.

Public: visible to ALL users of the application.

Developers can create 'Primary' (default) saved reports.

Analytical SQL

Q108

INTERMEDIATE

ANALYTICAL SQL

What is a Window/Analytic Function in Oracle? Write a RANK() example.**A N S W E R**

Analytic functions calculate across a set of rows related to the current row without collapsing rows like GROUP BY does. They use the OVER() clause.

E X A M P L E

```
SELECT ENAME, SAL, DEPTNO,  
       RANK() OVER (  
         PARTITION BY DEPTNO  
         ORDER BY SAL DESC  
       ) AS SALARY_RANK_IN_DEPT  
FROM EMP;
```

Q109

INTERMEDIATE

ANALYTICAL SQL

What is the difference between RANK(), DENSE_RANK(), and ROW_NUMBER()?**A N S W E R**

RANK(): Ties get same rank, then SKIPS next number → 1, 1, 3.

DENSE_RANK(): Ties get same rank, does NOT skip → 1, 1, 2.

ROW_NUMBER(): Always unique sequential, even for ties → 1, 2, 3.

E X A M P L E

```
-- Salaries: 5000, 5000, 3000  
SELECT SAL,
```

```
RANK() OVER (ORDER BY SAL DESC) AS RNK,  
DENSE_RANK() OVER (ORDER BY SAL DESC) AS DENSE_RNK,  
ROW_NUMBER() OVER (ORDER BY SAL DESC) AS ROW_NUM  
FROM EMP;  
-- Result: 5000→1,1,1 | 5000→1,1,2 | 3000→3,2,3
```

Q110 INTERMEDIATE

ANALYTICAL SQL

What is a WITH clause (CTE) in Oracle? When to use over subquery?**A N S W E R**

WITH clause (Common Table Expression) defines a named temporary result set referenced multiple times in the main query. Use when: (1) same subquery needed more than once, (2) complex query needs readable decomposition, (3) recursive queries needed.

E X A M P L E

```
WITH DEPT_AVG AS (  
  SELECT DEPTNO, AVG(SAL) AS AVG_SAL  
  FROM EMP GROUP BY DEPTNO  
)  
SELECT E.ENAME, E.SAL, D.AVG_SAL  
FROM EMP E  
JOIN DEPT_AVG D ON E.DEPTNO = D.DEPTNO  
WHERE E.SAL > D.AVG_SAL;
```

DML Advanced**Q111** INTERMEDIATE

DML ADVANCED

What is a MERGE statement in Oracle? Write a real example.**A N S W E R**

MERGE (UPSERT) inserts a row if it does not exist, or updates it if it does — in a single statement. Replaces separate INSERT + UPDATE conditional logic.

E X A M P L E

```
MERGE INTO CUSTOMERS C  
USING (SELECT 101 AS ID, 'Ravi' AS NAME FROM DUAL) SRC  
ON (C.CUST_ID = SRC.ID)  
WHEN MATCHED THEN  
  UPDATE SET C.CUST_NAME = SRC.NAME, UPDATED_ON = SYSDATE  
WHEN NOT MATCHED THEN  
  INSERT (CUST_ID, CUST_NAME, CREATED_ON)  
  VALUES (SRC.ID, SRC.NAME, SYSDATE);
```

Hierarchical SQL

Q112

INTERMEDIATE

HIERARCHICAL SQL

What is CONNECT BY in Oracle? Write a hierarchical query example.

A N S W E R

CONNECT BY is Oracle's syntax for querying hierarchical (tree-structured) data such as org charts, category trees. START WITH defines the root, CONNECT BY defines the parent-child relationship.

E X A M P L E

```
SELECT EMPNO, ENAME, MGR, LEVEL,  
       LPAD(' ', 2*(LEVEL-1)) || ENAME AS ORG_CHART  
FROM EMP  
START WITH MGR IS NULL  
CONNECT BY PRIOR EMPNO = MGR  
ORDER SIBLINGS BY ENAME;
```

Performance

Q113

INTERMEDIATE

PERFORMANCE

How do you read and interpret an execution plan in Oracle?

A N S W E R

Run EXPLAIN PLAN FOR your_query then SELECT * FROM TABLE(DBMS_XPLAN.DISPLAY).

Key things: TABLE ACCESS FULL on large tables → needs index; INDEX RANGE SCAN → good; HIGH Cost rows → bottleneck.

E X A M P L E

```
EXPLAIN PLAN FOR  
  SELECT * FROM ORDERS WHERE CUST_ID = 101;  
SELECT * FROM TABLE(DBMS_XPLAN.DISPLAY);  
  
-- TABLE ACCESS FULL (bad on large tables) → add index  
-- INDEX RANGE SCAN (good) → index is being used
```


Indexes

Q114

INTERMEDIATE

INDEXES

What is an index in Oracle? When does Oracle NOT use an index?

ANSWER

An index speeds up data retrieval. Oracle may NOT use an index when:

- (1) Function applied to indexed column: WHERE UPPER(EMAIL) = 'X'.
- (2) Not selective enough (returns > 10-15% of rows).
- (3) Table is very small.
- (4) NULL values — B-tree indexes do not index NULLs.

EXAMPLE

-- Index NOT used (function on column):

```
SELECT * FROM EMP WHERE SAL + 100 > 5000;
```

-- Index USED:

```
SELECT * FROM EMP WHERE SAL > 4900;
```

-- Fix with function-based index:

```
CREATE INDEX IDX_UPPER_EMAIL ON CUSTOMERS(UPPER(EMAIL));
```

DML Advanced

Q115

INTERMEDIATE

DML ADVANCED

How do you find and delete duplicate records from a table?

ANSWER

Find duplicates using GROUP BY + HAVING. Delete keeping the lowest ROWID (oldest row), remove all others.

EXAMPLE

-- Find duplicates:

```
SELECT EMAIL, COUNT(*) FROM CUSTOMERS  
GROUP BY EMAIL HAVING COUNT(*) > 1;
```

-- Delete duplicates, keep one row per email:

```
DELETE FROM CUSTOMERS  
WHERE ROWID NOT IN (  
  SELECT MIN(ROWID) FROM CUSTOMERS GROUP BY EMAIL  
);
```

SQL Functions**Q116** INTERMEDIATE

SQL FUNCTIONS

What is LISTAGG in Oracle? Write an example.**A N S W E R**

LISTAGG is an Oracle aggregate function that concatenates values from multiple rows into a single delimited string. Commonly used to show comma-separated lists in reports.

E X A M P L E

```
SELECT DEPTNO,  
       LISTAGG(ENAME, ', ')  
         WITHIN GROUP (ORDER BY ENAME) AS EMPLOYEES  
FROM EMP  
GROUP BY DEPTNO;  
-- Output:  
-- 10 | CLARK, KING, MILLER
```

Q117 INTERMEDIATE

SQL FUNCTIONS

What is CASE WHEN vs DECODE in Oracle SQL?**A N S W E R**

DECODE: Oracle-specific. Simple equality comparison only. Less readable.

CASE WHEN: ANSI standard. More readable. Supports complex conditions (ranges, LIKE, IS NULL). Preferred in modern Oracle.

E X A M P L E

```
-- DECODE (Oracle-specific, equality only):  
SELECT DECODE(DEPTNO,10,'Accounting',20,'Research','Other')  
FROM EMP;  
  
-- CASE WHEN (preferred, supports ranges):  
SELECT CASE  
  WHEN SAL >= 5000 THEN 'Senior'  
  WHEN SAL >= 3000 THEN 'Mid-level'  
  ELSE 'Junior'  
END AS GRADE FROM EMP;
```

Date Functions

Q118 INTERMEDIATE

DATE FUNCTIONS

How do you handle date arithmetic in Oracle SQL?

ANSWER

Oracle DATE type stores date and time. Adding/subtracting integers adds/subtracts DAYS. For months use ADD_MONTHS(). For differences use MONTHS_BETWEEN().

EXAMPLE

```
SELECT
  SYSDATE + 7          AS NEXT_WEEK,
  SYSDATE - 1          AS YESTERDAY,
  ADD_MONTHS(SYSDATE, 3) AS THREE_MONTHS_LATER,
  MONTHS_BETWEEN(SYSDATE, HIREDATE) AS MONTHS_EMPLOYED,
  TRUNC(SYSDATE, 'MM')  AS FIRST_OF_MONTH
FROM EMP WHERE ROWNUM = 1;
```

PL/SQL Advanced

Q119 INTERMEDIATE

PL/SQL ADVANCED

What is a REF CURSOR in PL/SQL? When would you use it?

ANSWER

A REF CURSOR is a cursor variable that can be opened for any SQL query at runtime — not tied to a fixed query at declaration. Use when returning a result set from a procedure to the calling program (Java, APEX), or when the query is built dynamically.

EXAMPLE

```
CREATE OR REPLACE PROCEDURE get_emp_by_dept(
  p_deptno IN NUMBER,
  p_cursor OUT SYS_REFCURSOR
) IS
BEGIN
  OPEN p_cursor FOR
    SELECT ENAME, SAL FROM EMP
    WHERE DEPTNO = p_deptno
    ORDER BY SAL DESC;
END;
```

PL/SQL Performance**Q120** INTERMEDIATE

PL/SQL PERFORMANCE

What is BULK COLLECT and FORALL in PL/SQL? Why use them?**A N S W E R**

BULK COLLECT: Fetches multiple rows from a cursor into a collection in ONE database call. Avoids row-by-row context switching.

FORALL: Executes DML for multiple rows in ONE database call.

Together they can be 10x–100x faster than row-by-row loops.

E X A M P L E

```
DECLARE
  TYPE t_emp IS TABLE OF EMP%ROWTYPE;
  v_emps t_emp;
BEGIN
  SELECT * BULK COLLECT INTO v_emps
  FROM EMP WHERE DEPTNO = 10;

  FORALL i IN v_emps.FIRST..v_emps.LAST
    UPDATE EMP SET SAL = SAL * 1.1
    WHERE EMPNO = v_emps(i).EMPNO;
END;
```

PL/SQL Advanced**Q121** INTERMEDIATE

PL/SQL ADVANCED

What is PRAGMA AUTONOMOUS_TRANSACTION? Give a real use case.**A N S W E R**

Makes a PL/SQL block run in its own independent transaction. Changes can be committed or rolled back independently of the calling transaction. Key use case: audit logging — log an error even if the main transaction rolls back.

E X A M P L E

```
CREATE OR REPLACE PROCEDURE log_error(
  p_msg VARCHAR2
) IS
  PRAGMA AUTONOMOUS_TRANSACTION;
BEGIN
  INSERT INTO ERROR_LOG(MSG, LOG_USER, LOG_TIME)
  VALUES(p_msg, :APP_USER, SYSDATE);
  COMMIT; -- commits ONLY this log, not main transaction
END;
```

Q122**INTERMEDIATE****PL/SQL ADVANCED**

What is EXECUTE IMMEDIATE in PL/SQL? When is it needed?

ANSWER

EXECUTE IMMEDIATE runs a SQL statement built as a string at runtime. Needed when table name, column name, or the entire query is not known until runtime. Always use bind variables — never concatenate user input directly.

EXAMPLE

```
DECLARE
  v_table VARCHAR2(30) := 'PRODUCTS';
  v_count NUMBER;
BEGIN
  EXECUTE IMMEDIATE
    'SELECT COUNT(*) FROM ' ||
    DBMS_ASSERT.SIMPLE_SQL_NAME(v_table) ||
    ' WHERE STATUS = :s'
    INTO v_count
    USING 'ACTIVE';
END;
```

Exception Handling

Q123**INTERMEDIATE****EXCEPTION HANDLING**

What is the difference between RAISE and RAISE_APPLICATION_ERROR?

ANSWER

RAISE: Re-raises the current exception or raises a predefined Oracle exception.

RAISE_APPLICATION_ERROR: Creates a custom error with your own message and error number (-20000 to -20999). The custom message is returned to the caller.

EXAMPLE

```
-- RAISE: predefined exception
RAISE NO_DATA_FOUND;

-- RAISE_APPLICATION_ERROR: custom message
IF v_count = 0 THEN
  RAISE_APPLICATION_ERROR(
    -20001,
    'Customer not found with ID: ' || p_id
  );
END IF;
```

APEX JSON**Q124****INTERMEDIATE****APEX JSON****What is APEX_JSON in APEX? Give 3 use cases.****A N S W E R**

APEX_JSON is the APEX package for building and parsing JSON in PL/SQL.

Use cases:

1. Build JSON response in an AJAX Callback.
2. Parse a JSON response from an external REST API.
3. Build JSON request body for POST calls.

E X A M P L E

```
-- BUILD JSON in AJAX Callback:
APEX_JSON.OPEN_OBJECT;
APEX_JSON.WRITE('status', 'OK');
APEX_JSON.WRITE('price', 299.99);
APEX_JSON.WRITE('stock', 45);
APEX_JSON.CLOSE_OBJECT;
-- Output: {"status":"OK","price":299.99,"stock":45}
```

Transactions**Q125****INTERMEDIATE****TRANSACTIONS****What is the difference between COMMIT, ROLLBACK, and SAVEPOINT?****A N S W E R**

COMMIT: Permanently saves all DML changes in the current transaction.

ROLLBACK: Undoes all DML changes since the last COMMIT.

SAVEPOINT name: Marks a point in the transaction. ROLLBACK TO name undoes only back to that point.

E X A M P L E

```
BEGIN
  INSERT INTO EMP VALUES(999, 'Test');
  SAVEPOINT after_insert;

  UPDATE EMP SET SAL = -1 WHERE EMPNO = 999; -- mistake!
  ROLLBACK TO after_insert; -- undo UPDATE only
  -- INSERT is still pending
```

```
COMMIT; -- permanently save the INSERT  
END;
```

JavaScript in APEX

Q126

INTERMEDIATE

JAVASCRIPT IN APEX

How do you get and set the value of an APEX page item using JavaScript?

ANSWER

Use `apex.item(itemName).getValue()` and `setValue()`. This is the correct APEX API — it works for all item types (select lists, date pickers, radio groups).

EXAMPLE

```
// GET value:  
var custId = apex.item('P3_CUST_ID').getValue();  
var name = apex.item('P3_NAME').getValue();  
  
// SET value:  
apex.item('P3_STATUS').setValue('ACTIVE');  
// For select list (return value, display value):  
apex.item('P3_DEPT_ID').setValue('10', 'Accounting');  
  
// Shorthand:  
var val = $v('P3_DEPT_ID');  
$s('P3_STATUS', 'ACTIVE');
```

Q127

INTERMEDIATE

JAVASCRIPT IN APEX

What is `apex.server.process()` and when do you use it?

ANSWER

`apex.server.process()` calls an APEX AJAX Callback from custom JavaScript. Use it when you need full control over the AJAX call, send specific parameters, or process the JSON response in JS before updating the UI.

EXAMPLE

```
apex.server.process(  
  'CALCULATE_TOTALS',  
  {  
    pageItems: '#P3_QTY,#P3_UNIT_PRICE',  
    x01: 'INCLUDE_TAX'  
  },  
  {  
    success: function(data) {  
      apex.item('P3_TOTAL').setValue(data.total);  
      apex.message.showPageSuccess('Calculated!');
```

```
    },  
    error: function(err) { console.error('AJAX error:', err); }  
  }  
);
```

Q128 INTERMEDIATE

JAVASCRIPT IN APEX

What is the apex namespace in APEX? Name 5 objects inside it.

A N S W E R

apex is the global JavaScript namespace for all Oracle APEX JavaScript APIs.

E X A M P L E

```
apex.item('P3_NAME').getValue() // page item operations  
apex.region('myGrid').refresh() // region operations  
apex.server.process('PROC', ...) // AJAX calls  
apex.message.showPageSuccess('OK!') // UI notifications  
apex.navigation.redirect('f?p=...') // page navigation  
apex.debug.log('val:', val) // debug logging  
apex.event.trigger(ctx, 'myEvent') // custom events
```

Q129 INTERMEDIATE

JAVASCRIPT IN APEX

How do you trigger a Dynamic Action from JavaScript in APEX?

A N S W E R

Fire a custom APEX event from JavaScript using `apex.event.trigger()`. Create a Dynamic Action that listens for that custom event (Event = Custom, Custom Event = your event name).

E X A M P L E

```
// Fire custom event:  
apex.event.trigger(  
  apex.gPageContext$,  
  'openAIAgent',  
  { prodId: 101 }  
);  
  
// Dynamic Action setup:  
// Event = Custom  
// Custom Event = openAIAgent  
// Selection Type = JavaScript Expression  
// Expression = apex.gPageContext$
```


CSS in APEX

Q130 INTERMEDIATE

CSS IN APEX

What is the difference between inline, page-level, and application-level CSS in APEX?

A N S W E R

Inline CSS: In HTML Attributes or style property. Scope = that one component. Highest specificity.

Page-level CSS: Page Properties → CSS → Inline. Scope = that page only.

Application-level CSS: Uploaded to Static Application Files, referenced in Theme. Scope = ALL pages.

E X A M P L E

```
:root { --brand-teal: #0d9488; } /* app-level: all pages */

/* Page-level (Page Properties → CSS → Inline) */
.my-form-region { box-shadow: 0 2px 8px rgba(0,0,0,0.1); }

/* Inline (Region HTML Attributes) */
/* style='border-radius:12px;' */
```

CSS Layout

Q131 INTERMEDIATE

CSS LAYOUT

What is Flexbox and when would you use it in APEX custom CSS?

A N S W E R

Flexbox is a CSS layout model for arranging elements in a row or column with flexible sizing. In APEX: align buttons side-by-side, create responsive card layouts, align form labels horizontally.

E X A M P L E

```
.apex-button-row {
  display: flex;
  gap: 12px;
  justify-content: flex-end; /* right-align buttons */
}

/* Responsive card grid */
.product-cards {
  display: flex;
  flex-wrap: wrap;
  gap: 16px;
}
.product-card { flex: 1 1 300px; }
```

CSS in APEX

Q132 INTERMEDIATE

CSS IN APEX

What are CSS custom properties (variables) and how do you use them in APEX themes?

ANSWER

CSS custom properties (variables) are defined with -- prefix on :root and referenced with var(). Universal Theme uses them extensively. Define brand colours as variables — then changing a colour is a one-line update.

EXAMPLE

```
:root {
  --vdx-primary: #0d9488; /* teal brand */
  --vdx-accent: #f97316; /* orange accent */
  --vdx-surface: #f8faff; /* card background */
  --vdx-border: #e2e8f0; /* border */
  --vdx-radius: 8px; /* border radius */
}

.my-card {
  background: var(--vdx-surface);
  border: 1px solid var(--vdx-border);
  border-radius: var(--vdx-radius);
}
```

JavaScript

Q133 INTERMEDIATE

JAVASCRIPT

What is the difference between == and === in JavaScript?

ANSWER

== (loose equality): Compares values AFTER type conversion. '5' == 5 → TRUE.

=== (strict equality): Compares values AND type WITHOUT conversion. '5' === 5 → FALSE.

Always use === in APEX JavaScript to avoid unexpected type coercion bugs.

EXAMPLE

```
'5' == 5 // true — type coerced
'5' === 5 // false — different types
0 == false // true — coerced
0 === false // false — different types
```

```
// In APEX:
if (apex.item('P3_QTY').getValue() === '') {
```

```
apex.message.showErrors('Quantity is required');
}
```

JavaScript in APEX

Q134 INTERMEDIATE

JAVASCRIPT IN APEX

How do you debug JavaScript errors in an APEX application?

ANSWER

1. Browser DevTools (F12) → Console tab: JS errors with file and line numbers.
2. Network tab: check AJAX calls — inspect request/response.
3. Enable APEX Debug Mode: server-side timing and process info.
4. Use `apex.debug.log()` in JS instead of `console.log`.
5. Add debugger; statement to pause execution.

EXAMPLE

```
apex.debug.log('Before AJAX:',
  apex.item('P3_CUST_ID').getValue());

apex.server.process('LOAD_DATA', { pagemItems:'#P3_CUST_ID' }, {
  success: function(data) {
    apex.debug.log('Response:', data);
    apex.item('P3_RESULT').set_value(data.value);
  },
  error: function(err) {
    apex.debug.error('AJAX failed:', err);
  }
});
```

Templates

Q135 INTERMEDIATE

TEMPLATES

What is template directive syntax in APEX? Give 2 real examples.

ANSWER

Template directives are special tags inside APEX template components for conditional logic, loops, and substitutions.

EXAMPLE

-- Conditional rendering:

```
{if STATUS_CD /}  
  <span class='status-badge status-{STATUS_CD/}'>  
    {STATUS_LABEL/}  
  </span>  
{endif/}
```

-- Substitution in template:

```
<div class='customer-card'>  
  <h3>{CUST_NAME/}</h3>  
  <p>Account: {ACCOUNT_NO/}</p>  
  <span class='balance'>{BALANCE/}</span>  
</div>
```

APEX Architecture

Q136 EXPERIENCED

APEX ARCHITECTURE

Explain the complete APEX page request-response lifecycle.

ANSWER

1. User navigates to URL (f?p=APP:PAGE:SESSION).
2. ORDS receives the HTTP request.
3. ORDS routes to APEX Engine inside Oracle Database.
4. APEX Engine: reads metadata, runs Before Header processes, applies Authentication/Authorization.
5. Renders HTML: executes region SQL queries, assembles templates.
6. Returns HTML → ORDS → Browser.
7. On Submit: APEX runs Validations → Processes → Branches.

EXAMPLE

Browser → ORDS (HTTP listener) → APEX Engine (in DB)

- ├─ Authentication check
- ├─ Before Header: Computations, Processes
- ├─ Region SQL execution → template assembly
- └─ HTML response → ORDS → Browser

On Submit:

- └─ Validations → Processes → Branches → Next Page

Q137 EXPERIENCED

APEX ARCHITECTURE

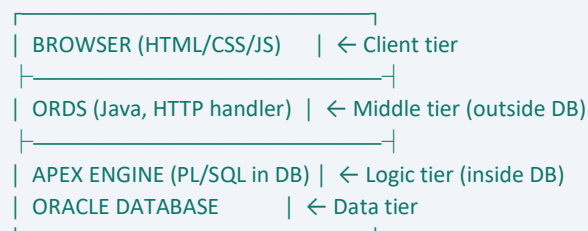
What is the difference between APEX Engine and ORDS in request handling?

ANSWER

APEX Engine: PL/SQL code inside Oracle Database. Reads APEX metadata, runs application logic, generates HTML. It IS the application server — but runs in the DB tier.

ORDS: Java application running OUTSIDE the database. Handles HTTP/HTTPS, manages DB connection pool, routes requests to APEX Engine.

EXAMPLE



Architecture Design

Q138 EXPERIENCED

ARCHITECTURE DESIGN

How would you design a multi-tenant APEX application with row-level isolation?

ANSWER

Row-level isolation (recommended for scale): Single schema, all tables have TENANT_ID column. Apply VPD policy at the DB level — WHERE clause is added automatically to every query.

Application Item: Store TENANT_ID in G_TENANT_ID at login. All SQL queries filter by :G_TENANT_ID.

EXAMPLE

```
CREATE FUNCTION tenant_filter(  
  p_schema VARCHAR2, p_table VARCHAR2  
) RETURN VARCHAR2 IS  
BEGIN  
  RETURN 'TENANT_ID = ''' ||  
    SYS_CONTEXT('APEX$SESSION','APP_USER') || ''';  
END;  
  
DBMS_RLS.ADD_POLICY(  
  object_name => 'CUSTOMERS',  
  policy_function => 'TENANT_FILTER'  
);  
-- Every query on CUSTOMERS auto-filters by TENANT_ID
```

DevOps & CI/CD

Q139 EXPERIENCED

DEVOPS & CI/CD

How do you implement CI/CD for an APEX application using VB Studio?

ANSWER

1. Export APEX app as SQL or APEXlang files.
2. Commit to Git in Oracle Visual Builder Studio.
3. Create Pipeline: Build (LoadCodeApp), Test, Deploy stages.
4. On merge request approval, pipeline auto-deploys to target environment.
5. APEXlang format makes Git diffs meaningful — see exactly which page changed.

EXAMPLE

VBS Pipeline:

```
1. Build → LoadCodeApp (imports APEX SQL) |  
2. Test → Automated test suite |  
3. Stage → Deploy to UAT environment |  
4. Approve → Manual gate (Team Lead) |  
5. Prod → AppPipelineSTG → deploy |
```

APEX 26.1 Features

Q140**EXPERIENCED****APEX 26.1 FEATURES**

What is APEXlang in APEX 26.1? Why does it matter for version control and AI?

ANSWER

APEXlang is an open, declarative, human-readable specification language for APEX applications. Apps export as .apx text files (one per page) instead of a single SQL blob.

Why it matters:

1. Git-friendly — meaningful diffs show exactly what changed on which page.
2. AI-ready — LLMs can read, modify, and generate APEXlang.
3. Validated by SQLcl before import.
4. Page Designer has a live APEXlang View for any page.

EXAMPLE

-- Simplified APEXlang page structure:

```
page 3 (  
  title: "Customer Form",  
  alias: "customers",  
  items: [  
    item P3_CUST_ID ( type: hidden ),  
    item P3_CUST_NAME ( type: text_field, label: "Name" )  
  ],  
  buttons: [ button SAVE ( action: submit ) ]  
)
```

Q141**EXPERIENCED****APEX 26.1 FEATURES**

What JavaScript property names changed in APEX 26.1 that break existing Dynamic Actions?

ANSWER

In APEX 26.1, inside Dynamic Actions at the page-level JavaScript context, two property names were renamed:

- regionStaticId → regionDomId
- staticId → domId

Any existing DA using this.regionStaticId or this.staticId will silently break after upgrade. Oracle provides a diagnostic SQL in the 26.1 release notes to find all affected DAs.

EXAMPLE

```
-- Run BEFORE upgrading to 26.1:  
SELECT page_id, component_name, attribute_01  
FROM apex_application_page_da_acts  
WHERE application_id = :APP_ID
```

```
AND (attribute_01 LIKE '%regionStaticId%'
     OR attribute_01 LIKE '%this.staticId%');
-- Fix: replace with regionDomId / domId
```

APEX AI

Q142 EXPERIENCED

APEX AI

What is the APEX_AI package in APEX 26.1? Name 3 key functions.

ANSWER

APEX_AI is the PL/SQL package for interacting with Generative AI services.

Key functions:

1. APEX_AI.GENERATE(p_prompt, p_service_static_id) — generate text from a service.
2. APEX_AI.CHAT(p_config_static_id, p_prompt) — conversational chat with history.
3. APEX_AI.SET_TOOL_RESULT(p_result) — mandatory in AI Agent tools to return data to the AI loop.

EXAMPLE

```
DECLARE l_summary CLOB;
BEGIN
  l_summary := APEX_AI.GENERATE(
    p_prompt      => 'Summarize in 2 sentences: ' || :P3_NOTES,
    p_service_static_id => 'router_cred'
  );
  :P3_AI_SUMMARY := l_summary;
END;
```

Q143 EXPERIENCED

APEX AI

What is a Generative AI Service in APEX 26.1? What are the required configuration fields?

ANSWER

A Generative AI Service stores the connection config for an AI provider (Shared Component).

Required: (1) AI Provider, (2) Base URL, (3) Web Credential, (4) Model name, (5) Static ID.

Critical: Leave Maximum AI Tokens BLANK for OpenRouter — any value causes ORA-20962.

EXAMPLE

```
Provider:   OpenAI-Compatible (HTTP)
Base URL:   https://openrouter.ai/api/v1
Credential: AI_POWERED_CRED
Model:      google/gemini-2.5-flash-lite-preview-09-2025
Static ID:   router_cred
Max Tokens: [LEAVE BLANK → avoids ORA-20962]
```


HTTP Headers:
HTTP-Referer: <https://apex.oracle.com>
X-Title: MyAPEXApp

Q144 EXPERIENCED

APEX AI

What is an AI Agent in Oracle APEX 26.1? How is it different from a chatbot?**A N S W E R**

AI Agent in APEX 26.1 is a reasoning engine that can TAKE ACTION inside your application through approved AI Tools.

Chatbot: Generates text responses. Passive. Cannot read DB data or update anything.

AI Agent: Can CALL TOOLS — read from your database (SQL query tools), execute PL/SQL (update data, create records), trigger JavaScript.

E X A M P L E

```
-- Vendix Stock Manager AI Agent:
-- User: 'How many Samsung TVs are in stock?'
-- Agent calls Tool 1 (Retrieve Data SQL):
SELECT PROD_NAME, STOCK_QTY FROM POS_INVENTORY
WHERE CATEGORY = :CATEGORY
-- Returns: Samsung 32 Inch HD LED TV - 10 units

-- User: 'Update Samsung TV stock to 50'
-- Agent calls Tool 2 (Execute Server-side Code PL/SQL)
-- Updates POS_INVENTORY + logs to POS_STOCK_MOVEMENTS
```

Q145 EXPERIENCED

APEX AI

What are the two types of AI Agent Tools — Augment System Prompt vs On Demand?**A N S W E R**

Augment System Prompt: Run AUTOMATICALLY on EVERY message. Output injected into system prompt as context. Used for: current user info, today's date, user role.

On Demand: Run ONLY when the AI model decides they are needed. Used for: database queries, PL/SQL actions, RAG data retrieval.

E X A M P L E

```
-- Augment System Prompt Tool (runs every message):
SELECT 'User: '||:APP_USER||
      ', Role: '||:G_USER_ROLE||
      ', Date: '||TO_CHAR(SYSDATE,'DD-MON-YYYY') FROM DUAL
-- Injected automatically into every AI message

-- On Demand Tool (AI calls only when needed):
-- CHECK_STOCK_BY_CATEGORY: called when user asks about stock
```

Q146 EXPERIENCED

APEX AI

What is APEX_AI.SET_TOOL_RESULT and why is it mandatory in Execute Server-side Code tools?

ANSWER

APEX_AI.SET_TOOL_RESULT(p_result CLOB) sends the tool execution result BACK to the AI model. Without it, the AI agent loop cannot continue — the model never gets the output and cannot generate a final response. Always call it at the end of every Execute Server-side Code tool.

EXAMPLE

```
DECLARE v_old_qty NUMBER;
BEGIN
  SELECT STOCK_QTY INTO v_old_qty
  FROM POS_INVENTORY WHERE PROD_ID = :PRODUCT_ID;

  UPDATE POS_INVENTORY
  SET STOCK_QTY = :QUANTITY
  WHERE PROD_ID = :PRODUCT_ID;

  INSERT INTO POS_STOCK_MOVEMENTS(PROD_ID,OLD_QTY,NEW_QTY,MOVED_BY)
  VALUES(:PRODUCT_ID, v_old_qty, :QUANTITY, :APP_USER);

  COMMIT;

  -- MANDATORY: send result back to AI loop
  APEX_AI.SET_TOOL_RESULT(
    'Updated: ' || v_old_qty || ' → ' || :QUANTITY || ' units.'
  );
END;
```

Q147 EXPERIENCED

APEX AI

What is NL2IR in APEX 26.1? How does APEX avoid SQL injection risk?

ANSWER

NL2IR lets end-users type natural language queries into an Interactive Report and APEX filters/groups the report automatically.

Security: APEX does NOT execute AI-generated SQL. The AI applies DECLARATIVE Interactive Report filter chips. The underlying SQL never changes.

EXAMPLE

```
-- User types: 'Show only ACTIVE customers in Delhi'

-- APEX adds declarative chips:
-- Filter: STATUS = 'ACTIVE'
-- Filter: CITY = 'Delhi'

-- APEX does NOT:
-- - Generate or execute new SQL
-- - Modify the base IR query
-- Each chip is visible and removable by the user
```

Security

Q148 EXPERIENCED

SECURITY

What is a Web Credential vs storing an API key in an Application Item?

ANSWER

Web Credential: Stores API keys ENCRYPTED in the APEX repository. Never visible in exports, code, or session state. Referenced by Static ID only.

Application Item: Stores values in session state (unencrypted). Can be seen in debug mode, accidentally logged.

Rule: ALWAYS use Web Credentials for API keys.

EXAMPLE

-- WEB CREDENTIAL (SECURE):

Type: HTTP Header

Name: Authorization

Secret: Bearer sk-openrouter-key ← encrypted at rest

ID: AI_POWERED_CRED

-- NEVER DO THIS:

G_API_KEY := 'sk-openrouter-key';

-- Visible in debug mode, session state, error logs!

APEX AI

Q149 EXPERIENCED

APEX AI

What is RAG (Retrieval-Augmented Generation) in APEX AI Configurations?

ANSWER

RAG fetches relevant data from your own database and includes it in the AI prompt before generating a response. Grounds the AI in your specific business data — not generic training data. Implemented via Augment System Prompt tools or AI Vector Search.

EXAMPLE

-- RAG flow:

-- 1. User: 'What is our return policy for electronics?'

-- 2. Augment System Prompt Tool runs:

```
SELECT POLICY_TEXT FROM COMPANY_POLICIES  
WHERE CATEGORY = 'RETURNS'
```

-- 3. AI receives:

-- System prompt + policy text + user question

-- 4. AI generates accurate answer based on YOUR data

APEX 26.1 Features

Q150

EXPERIENCED

APEX 26.1 FEATURES

What is Data Reporter in APEX 26.1? How does it differ from Interactive Reports?

ANSWER

Data Reporter is a new 26.1 self-service reporting feature for business users. Admin defines trusted datasets (approved SQL views). Users create their own reports via drag-and-drop. Published reports are shareable by URL.

Difference: IR requires a developer to build the page first. Data Reporter is fully user-built, admin-governed.

EXAMPLE

-- Admin (developer) creates approved dataset once:

```
CREATE OR REPLACE VIEW V_SALES_RPT AS
SELECT S.SALE_ID, S.SALE_DATE, C.CUST_NAME,
       P.PROD_NAME, S.QUANTITY, S.TOTAL_AMOUNT
FROM SALES S JOIN CUSTOMERS C ON S.CUST_ID = C.CUST_ID
JOIN PRODUCTS P ON S.PROD_ID = P.PROD_ID;
```

-- Business users explore it in Data Reporter

-- Zero developer time for each new report

Q151

EXPERIENCED

APEX 26.1 FEATURES

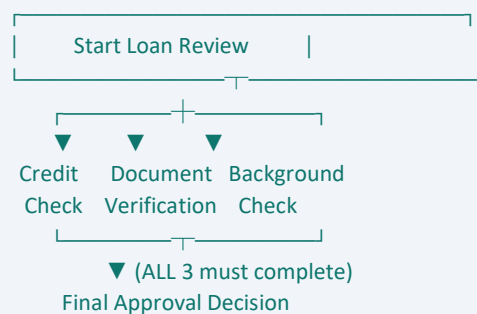
What is parallel flow support in APEX 26.1 Workflow? Give a use case.

ANSWER

APEX 26.1 added Parallel Flows to the Workflow engine. Multiple branches execute simultaneously and the workflow waits for ALL branches to complete before continuing.

EXAMPLE

-- Loan Approval (26.1 Parallel Flow):



Q152

EXPERIENCED

APEX 26.1 FEATURES

What changed in Interactive Grid clipboard support in APEX 26.1?

ANSWER

APEX 26.1 enhanced Interactive Grid clipboard operations:

- Copy selected cells (Ctrl+C)
- Cut selected cells (Ctrl+X)
- Paste cells (Ctrl+V)
- Paste Insert — insert new rows from clipboard (Shift+Ctrl+V)
- Drag-and-drop tabular data from Excel directly into the grid

Makes IG feel much closer to a spreadsheet for data entry.

Performance Tuning

Q153

EXPERIENCED

PERFORMANCE TUNING

How do you identify and fix a slow-running Interactive Report query in production?

ANSWER

1. Enable APEX Debug → view report render time and SQL execution time.
2. Copy IR base SQL → EXPLAIN PLAN in SQL Developer.
3. Look for TABLE ACCESS FULL on large tables → add indexes.
4. Check IR filters — bind variable issues.
5. Add IS NULL fallback in WHERE to avoid scanning all rows.
6. Consider Result Cache for infrequently-changing data.

EXAMPLE

```
EXPLAIN PLAN FOR
SELECT CUST_ID, CUST_NAME, STATUS
FROM CUSTOMERS
WHERE (:P3_STATUS IS NULL OR STATUS = :P3_STATUS)
ORDER BY CREATED_ON DESC;

SELECT * FROM TABLE(DBMS_XPLAN.DISPLAY);
-- Look for TABLE ACCESS FULL → add composite index:
CREATE INDEX IDX_CUST_STATUS
ON CUSTOMERS(STATUS, CREATED_ON, CUST_ID, CUST_NAME);
```

Q154**EXPERIENCED****PERFORMANCE TUNING**

What is Lazy Loading in APEX? When should you enable it?

ANSWER

Lazy Loading defers region rendering. The placeholder loads with the page, but the region's content loads in a separate async request after page load.

Enable when: region has a slow query, region is below the fold, or multiple regions exist and above-the-fold content should load first.

EXAMPLE

-- Enable:

-- Region → Advanced → Lazy Loading = Yes

-- What happens:

-- 1. Page loads immediately (fast, spinner shown)

-- 2. Region content loads 0.3-0.5s later separately

-- Best for: dashboard charts, large reports,

-- any region where a brief delay is acceptable

Q155**EXPERIENCED****PERFORMANCE TUNING**

How do you use bind variables effectively to prevent hard parsing in APEX?

ANSWER

Hard parsing compiles a new execution plan for every unique SQL string. Bind variables allow plan reuse (soft parse). In APEX: always use :P3_FILTER_VALUE in WHERE clause. Never concatenate user input — both a performance and security issue.

EXAMPLE

-- WRONG (hard parse + SQL injection risk):

-- Every value compiles a NEW execution plan:

'SELECT * FROM EMP WHERE DEPTNO = ' || :P3_DEPT

-- CORRECT (bind variable → plan reuse → safe):

SELECT EMPNO, ENAME, SAL

FROM EMP

WHERE (:P3_DEPT IS NULL OR DEPTNO = :P3_DEPT)

-- Same execution plan reused for all department values

Security

Q156 EXPERIENCED

SECURITY

What is XSS and how does APEX prevent it?

ANSWER

XSS is a security attack where malicious JavaScript is injected into a web page and executed in another user's browser. APEX prevents XSS by:

1. Automatically HTML-encoding all output from page items and report columns.
2. APEX_ESCAPE.HTML() for manual escaping in PL/SQL.
3. Column 'Escape Special Characters' = Yes by default in reports.

EXAMPLE

```
-- Manual escaping in PL/SQL Dynamic Content:
http.p('<div>' || APEX_ESCAPE.HTML(:P3_USER_COMMENT) || '</div>');

-- WITHOUT escape (XSS RISK - never do this):
http.p('<div>' || :P3_USER_COMMENT || '</div>');
-- If P3_USER_COMMENT = '<script>alert(1)</script>' -> EXECUTED!
```

Q157 EXPERIENCED

SECURITY

How do you implement row-level security using VPD in APEX?

ANSWER

VPD adds a WHERE clause to every SQL query on a table automatically at the DB level. Works below APEX — even SQL Developer and REST API queries are filtered. APEX reads the user via SYS_CONTEXT('APEX\$SESSION','APP_USER').

EXAMPLE

```
CREATE FUNCTION rls_filter(
  p_schema VARCHAR2, p_table VARCHAR2
) RETURN VARCHAR2 IS
BEGIN
  IF APEX_UTIL.IS_IN_GROUP('ADMIN') THEN
    RETURN NULL; -- admins see all rows
  END IF;
  RETURN 'ASSIGNED_TO = ''' ||
    SYS_CONTEXT('APEX$SESSION','APP_USER') || ''';
END;

DBMS_RLS.ADD_POLICY(
  object_name => 'CUSTOMERS',
  policy_function => 'RLS_FILTER'
);
```

Q158 EXPERIENCED

SECURITY

How do you implement a fully custom authentication scheme in APEX?

ANSWER

Go to Shared Components → Authentication Schemes → Create → Custom. Define the Credentials Verification Function — returns TRUE (valid) or FALSE (invalid). Add post-login setup code (e.g., set G_USER_ROLE).

EXAMPLE

```
CREATE OR REPLACE FUNCTION custom_auth(
  p_user IN VARCHAR2,
  p_pass IN VARCHAR2
) RETURN BOOLEAN IS
  v_count NUMBER;
BEGIN
  SELECT COUNT(*) INTO v_count
  FROM APP_USERS
  WHERE UPPER(USERNAME) = UPPER(p_user)
    AND PASSWORD_HASH = STANDARD_HASH(p_pass, 'SHA256')
    AND STATUS = 'ACTIVE';
  RETURN v_count > 0;
END;
```

Q159**EXPERIENCED****SECURITY**

How do you audit who accessed what in an APEX application?

ANSWER

1. Application audit: Page process logs APP_USER, page ID, action, timestamp to an AUDIT_LOG table.
2. Database triggers: AFTER DML on critical tables — log OLD/NEW values.
3. Use PRAGMA AUTONOMOUS_TRANSACTION so audit log persists even if main transaction rolls back.

EXAMPLE

```
CREATE OR REPLACE TRIGGER TRG_CUST_AUDIT
AFTER INSERT OR UPDATE OR DELETE ON CUSTOMERS
FOR EACH ROW
DECLARE PRAGMA AUTONOMOUS_TRANSACTION;
BEGIN
  INSERT INTO AUDIT_LOG(
    TABLE_NAME, OPERATION, PK_VALUE,
    OLD_VALUE, NEW_VALUE, CHANGED_BY, CHANGED_ON
  ) VALUES(
    'CUSTOMERS', ORA_SYSEVENT,
    NVL(:NEW.CUST_ID, :OLD.CUST_ID),
    :OLD.CUST_NAME, :NEW.CUST_NAME,
    SYS_CONTEXT('APEX$SESSION', 'APP_USER'), SYSDATE
  );
  COMMIT;
END;
```


REST & ORDS**Q160****EXPERIENCED****REST & ORDS****How do you create a REST endpoint using ORDS from within APEX?****A N S W E R**

Option 1: SQL Workshop → RESTful Services → Create Module, Template, Handler.

Option 2: Object Browser → right-click Table → Enable REST → ORDS auto-generates GET/POST/PUT/DELETE.

All options generate a REST endpoint at: `https://host/ords/schema/module/template`

E X A M P L E

-- Auto-generated endpoints after Enable REST:

```
GET /ords/vendix/customers/ -- get all
GET /ords/vendix/customers/:id -- get by ID
POST /ords/vendix/customers/ -- insert
PUT /ords/vendix/customers/:id -- update
DELETE /ords/vendix/customers/:id -- delete
```

REST & Integration**Q161****EXPERIENCED****REST & INTEGRATION****How do you handle large JSON payloads in PL/SQL without hitting VARCHAR2 limits?****A N S W E R**

VARCHAR2 is limited to 32,767 bytes in PL/SQL. For large JSON use CLOB data type.

APEX_WEB_SERVICE.MAKE_REST_REQUEST returns CLOB. APEX_JSON.PARSE() accepts CLOB.

E X A M P L E

```
DECLARE
  l_response CLOB; -- NOT VARCHAR2(32767)
  l_content CLOB;
BEGIN
  -- Large AI API response — use CLOB:
  l_response := APEX_WEB_SERVICE.MAKE_REST_REQUEST(
    p_url => 'https://openrouter.ai/api/v1/chat/completions',
    p_http_method => 'POST',
    p_body => l_request_clob,
    p_credential_static_id => 'AI_POWERED_CRED'
  );
  APEX_JSON.PARSE(l_response);
  l_content := APEX_JSON.GET_CLOB_VALUE(
    'choices[1].message.content');
END;
```

Q162

EXPERIENCED

REST & INTEGRATION

How do you configure an OpenRouter API integration in APEX?

ANSWER

1. Create Web Credential: HTTP Header, Authorization, Bearer YOUR_KEY, Static ID = AI_POWERED_CRED.
2. Create Generative AI Service: Provider = OpenAI-Compatible, Base URL = <https://openrouter.ai/api/v1>, leave Max AI Tokens BLANK.
3. Add required HTTP headers.

EXAMPLE

```
-- Web Credential:
Authentication Type: HTTP Header
Credential Name: Authorization
Credential Secret: Bearer sk-or-v1-your-openrouter-key
Static ID: AI_POWERED_CRED

-- Generative AI Service:
Provider: OpenAI-Compatible
Base URL: https://openrouter.ai/api/v1
Credential: AI_POWERED_CRED
Model: google/gemini-2.5-flash-lite-preview-09-2025
Static ID: router_cred
Max Tokens: [LEAVE BLANK → avoids ORA-20962]
```

Architecture Design

Q163

EXPERIENCED

ARCHITECTURE DESIGN

How do you switch AI providers without changing application code?

ANSWER

Design with a Static ID from day one. All APEX_AI.GENERATE calls and AI Agent configurations reference the Static ID only — never hardcode provider details in code. To switch: update ONLY the Generative AI Service shared component.

EXAMPLE

```
-- All code (never changes regardless of provider):
APEX_AI.GENERATE(
  p_prompt      => :P3_QUERY,
  p_service_static_id => 'MAIN_AI_SERVICE' -- never changes
);

-- To switch from OCI to OpenRouter:
-- Shared Components → Generative AI Services
-- → MAIN_AI_SERVICE → Edit → change Provider, URL, Model
-- Zero code changes anywhere in the application
```

PL/SQL Expert

Q164

EXPERIENCED

PL/SQL EXPERT

What is the difference between STANDARD_HASH and DBMS_CRYPTO? Which works on Oracle ADB?

ANSWER

STANDARD_HASH(value, algorithm): Built-in SQL/PL/SQL function for one-way hashing (SHA256, SHA512). Works on ADB without special grants. Simpler.

DBMS_CRYPTO: Full encryption/decryption package (AES, 3DES). Requires EXECUTE privilege grant from ADMIN on ADB.

For password hashing on ADB: use STANDARD_HASH.

EXAMPLE

```
-- STANDARD_HASH (works on ADB, no grants needed):
SELECT STANDARD_HASH('mypassword', 'SHA256') FROM DUAL;

-- Password validation:
SELECT COUNT(*) FROM APP_USERS
WHERE USERNAME = :P1_USER
AND PASSWORD_HASH = STANDARD_HASH(:P1_PASS, 'SHA256');

-- DBMS_CRYPTO (needs grant on ADB):
GRANT EXECUTE ON DBMS_CRYPTO TO your_schema;
```

Audit Trails

Q165

EXPERIENCED

AUDIT TRAILS

How do you implement a full audit trail for DML operations in an APEX application?

ANSWER

1. Add audit columns to all tables: CREATED_BY, CREATED_ON, UPDATED_BY, UPDATED_ON.
2. Create BEFORE INSERT/UPDATE triggers to populate audit columns.
3. Create a separate AUDIT_LOG table for full history (OLD/NEW values).
4. Use PRAGMA AUTONOMOUS_TRANSACTION so audit log persists even if main tx rolls back.

EXAMPLE

```
CREATE OR REPLACE TRIGGER TRG_CUST_AUDIT
BEFORE INSERT OR UPDATE ON CUSTOMERS
FOR EACH ROW
```

```
DECLARE
v_user VARCHAR2(100) :=
SYS_CONTEXT('APEX$SESSION','APP_USER');
BEGIN
IF INSERTING THEN
:NEW.CREATED_BY := v_user;
:NEW.CREATED_ON := SYSDATE;
END IF;
:NEW.UPDATED_BY := v_user;
:NEW.UPDATED_ON := SYSDATE;
END;
```

PL/SQL Expert

Q166**EXPERIENCED****PL/SQL EXPERT**

What is a COMPOUND trigger in Oracle and when is it needed?

ANSWER

A COMPOUND trigger combines multiple timing points (BEFORE/AFTER STATEMENT, BEFORE/AFTER EACH ROW) in one trigger body, sharing state between timing points. Needed to avoid the ORA-04091 mutating table error.

EXAMPLE

```
CREATE OR REPLACE TRIGGER TRG_EMP_COMPOUND
FOR INSERT OR UPDATE ON EMP
COMPOUND TRIGGER
TYPE t_ids IS TABLE OF NUMBER INDEX BY PLS_INTEGER;
g_changed_ids t_ids;
g_idx PLS_INTEGER := 0;

AFTER EACH ROW IS
BEGIN
g_idx := g_idx + 1;
g_changed_ids(g_idx) := :NEW.EMPNO;
END AFTER EACH ROW;

AFTER STATEMENT IS
BEGIN
FOR i IN 1..g_idx LOOP
recalculate_dept_stats(g_changed_ids(i));
END LOOP;
END AFTER STATEMENT;
END;
```

SQL Expert

Q167**EXPERIENCED****SQL EXPERT**

How do you implement pagination using OFFSET FETCH in Oracle SQL?

ANSWER

Modern Oracle (12c+) uses SQL-standard OFFSET/FETCH for pagination. Old approach used nested ROWNUM subqueries. APEX IR/IG handles pagination automatically — use OFFSET FETCH for custom paging in AJAX callbacks.

EXAMPLE

```
-- MODERN: Oracle 12c+ OFFSET FETCH:
SELECT PROD_ID, PROD_NAME, PRICE
FROM PRODUCTS
ORDER BY PROD_NAME
OFFSET 20 ROWS      -- skip first 20 (page 3)
FETCH NEXT 10 ROWS ONLY; -- get rows 21-30

-- LEGACY: ROWNUM nested subquery:
SELECT * FROM (
  SELECT p.*, ROWNUM rn FROM PRODUCTS p
  WHERE ROWNUM <= 30
) WHERE rn > 20;
```

Security

Q168**EXPERIENCED****SECURITY**

What is DBMS_ASSERT and why is it important in APEX security?

ANSWER

DBMS_ASSERT validates and sanitizes strings used in dynamic SQL. Prevents SQL injection when you must use EXECUTE IMMEDIATE with dynamic object names.

Key functions: SIMPLE_SQL_NAME (validates table/column name), ENQUOTE_LITERAL (safely quotes string).

EXAMPLE

```
-- SAFE dynamic table name:
EXECUTE IMMEDIATE
'SELECT COUNT(*) FROM ' ||
DBMS_ASSERT.SIMPLE_SQL_NAME(:P3_TABLE_NAME) ||
' WHERE STATUS = :s'
INTO v_count
USING 'ACTIVE';

-- DBMS_ASSERT raises ORA-44003 if input is invalid:
-- 'PRODUCTS; DROP TABLE ORDERS--' → ORA-44003
-- 'PRODUCTS' → accepted as valid SQL name
```

PL/SQL Expert

Q169

EXPERIENCED

PL/SQL EXPERT

What is the difference between SYS_CONTEXT and V() in APEX PL/SQL processes?**A N S W E R**

V('ITEM_NAME'): APEX function to retrieve session state. Only works in active APEX page request context.

SYS_CONTEXT('APEX\$SESSION','APP_USER'): Oracle DB function. Works inside triggers, VPD policies, and procedures called from non-APEX contexts (SQL Developer, REST API).

E X A M P L E

```
-- V() in APEX page process (correct for APEX context):
v_user := V('APP_USER');
v_dept := V('P3_DEPT_ID');

-- SYS_CONTEXT in DB trigger (no APEX session here):
CREATE OR REPLACE TRIGGER TRG_AUDIT
BEFORE UPDATE ON CUSTOMERS FOR EACH ROW
BEGIN
  :NEW.UPDATED_BY :=
    SYS_CONTEXT('APEX$SESSION','APP_USER');
  -- Can NOT use V() here — no active APEX page context
END;
```

Scenario: Debug

Q170

EXPERIENCED

SCENARIO: DEBUG

An Interactive Grid is not saving changes. Walk through your debugging checklist.**A N S W E R**

1. Check IG Attributes → Edit → Enabled = Yes.
2. Check IG SQL — must be a single-table query with a Primary Key column.
3. Check Row Primary Key — IG must know the PK column.
4. Enable APEX Debug → check for ERROR messages.
5. Remove all validations temporarily to isolate.
6. Check if Authorization Scheme is blocking the save.

E X A M P L E

```
-- Debugging steps:
-- 1. IG → Attributes → Edit → Enabled = Yes
-- 2. IG SQL: SELECT * FROM ORDERS ← single table
-- 3. IG → Columns → ORDER_ID → Primary Key = Yes
```

- 4. Enable APEX Debug: &p_debug=YES in URL
- Session → View Debug → look for ERROR messages
- 5. Remove all validations → test save → add back one by one

Q171 EXPERIENCED

SCENARIO: DEBUG

Your app works in DEV but throws ORA-01422 in PROD. What do you check?**A N S W E R**

ORA-01422 occurs when SELECT INTO finds multiple rows but expects exactly one. Root cause: DEV has limited test data but PROD has real data returning multiple rows.

Fix: Add ROWNUM = 1, use aggregate (MAX/MIN), add a more restrictive WHERE clause, or switch to a cursor loop.

E X A M P L E

```
-- ORA-01422 at runtime:
SELECT STATUS INTO v_status
FROM ORDERS WHERE CUST_ID = :P3_ID; -- multiple rows!
```

```
-- Fix: get most recent order:
SELECT STATUS INTO v_status
FROM ORDERS
WHERE CUST_ID = :P3_ID
ORDER BY ORDER_DATE DESC
FETCH FIRST 1 ROW ONLY;
```

Q172 EXPERIENCED

SCENARIO: DEBUG

A Dynamic Action set to fire on Page Load is not executing. Walk through debugging.**A N S W E R**

1. F12 → Console tab: check for JavaScript errors stopping execution.
2. Check DA condition — is it preventing the DA from firing?
3. Check if the affected region uses Lazy Loading — item may not exist when DA fires.
4. Enable APEX Debug → check if DA is listed in debug output.
5. Temporarily remove the condition to confirm the DA fires unconditionally.

E X A M P L E

```
-- Debugging:
-- 1. F12 → Console → any JavaScript errors?
-- 2. Check DA condition: is 'Item = Value' check failing?
-- 3. If region is Lazy Loaded:
--   DA fires before region content loads
--   Fix: change DA event to 'After Refresh' of that region
-- 4. APEX Debug → View Debug → search for DA name
-- 5. Temporarily set DA condition = 'No Condition'
```

Scenario: Architecture**Q173** EXPERIENCED

SCENARIO: ARCHITECTURE

You need to pass 50KB of JSON from an APEX page to a PL/SQL process. What do you do?

A N S W E R

VARCHAR2 session state is limited to 32,767 bytes. For large payloads:

1. Use a CLOB page item (Text Area with CLOB source).
2. Use apex.server.process and send data via clob01 — PL/SQL reads APEX_APPLICATION.G_CLOB_01.
3. Use APEX_COLLECTION with CLOB001 member.

E X A M P L E

```
// JavaScript: send large JSON via AJAX:
apex.server.process(
  'PROCESS_LARGE_JSON',
  {
    x01: 'metadata',
    clob01: JSON.stringify(largeDataObject) // 50KB here
  },
  { success: function(r) { ... } }
);

-- PL/SQL reads it:
DECLARE l_json CLOB := APEX_APPLICATION.G_CLOB_01;
BEGIN
  APEX_JSON.PARSE(l_json);
  -- process the 50KB JSON
END;
```

Scenario: Performance**Q174** EXPERIENCED

SCENARIO: PERFORMANCE

An APEX page has a 4-second load time in production. How do you profile and fix it?

A N S W E R

1. Enable APEX Debug → identify slow regions and processes.
2. Copy slow region SQL → EXPLAIN PLAN → look for Full Table Scans.
3. Add indexes on WHERE/ORDER BY columns.
4. Check cascading LOV refreshes on load — each queries DB.
5. Enable Lazy Loading on below-the-fold regions.
6. Check for N+1 problem — nested queries inside loops.

E X A M P L E

```
-- Debug output reveals:  
-- Region 'Dashboard KPIs': 2.3s ← slowest  
-- Region 'Recent Orders': 0.8s  
  
-- EXPLAIN PLAN shows:  
-- TABLE ACCESS FULL: SALES (3M rows!) ← root cause  
  
-- Fix:  
CREATE INDEX IDX_SALES_DATE  
  ON SALES(SALE_DATE, STATUS);  
  
-- AND: enable Lazy Loading on Recent Orders region
```

Scenario: Debug**Q175****EXPERIENCED****SCENARIO: DEBUG**

A learner reports HTTP 401 Unauthorized errors when calling the AI Agent. What are the 3 most likely root causes?

A N S W E R

1. Missing 'Bearer ' prefix — Authorization header must be 'Bearer YOUR_KEY' not just 'YOUR_KEY'. The space after 'Bearer' is mandatory. Most common cause.
2. Expired or invalid API key — check the provider dashboard for key status and credit balance.
3. Wrong credential referenced — Generative AI Service points to wrong Web Credential Static ID.

E X A M P L E

```
-- Root cause 1 (most common): missing 'Bearer ' prefix  
Web Credential → Credential Secret:  
WRONG: sk-or-v1-yourkey  
RIGHT: Bearer sk-or-v1-yourkey ← space after Bearer!  
  
-- Root cause 2: expired key  
-- Check: openrouter.ai → Keys → check expiry & balance  
  
-- Root cause 3: wrong Static ID  
-- Verify Generative AI Service → Credential matches exactly
```

Scenario: Architecture**Q176** EXPERIENCED

SCENARIO: ARCHITECTURE

How do you implement an AI Agent that reads AND updates inventory with a full audit trail?

A N S W E R

Create two AI Agent Tools:

Tool 1 (Retrieve Data): CHECK_STOCK — SQL query returning product name, category, stock qty.

Tool 2 (Execute Server-side Code): UPDATE_STOCK — PL/SQL that updates POS_INVENTORY, logs to POS_STOCK_MOVEMENTS, commits, then calls APEX_AI.SET_TOOL_RESULT.

E X A M P L E

```
-- Tool 1: Retrieve Data
SELECT PROD_NAME, CATEGORY, STOCK_QTY
FROM POS_INVENTORY
WHERE (:CATEGORY IS NULL OR CATEGORY = :CATEGORY)

-- Tool 2: Execute Server-side Code
DECLARE v_old NUMBER;
BEGIN
  SELECT STOCK_QTY INTO v_old
  FROM POS_INVENTORY WHERE PROD_ID = :PRODUCT_ID;

  UPDATE POS_INVENTORY SET STOCK_QTY = :QUANTITY
  WHERE PROD_ID = :PRODUCT_ID;

  INSERT INTO POS_STOCK_MOVEMENTS(
    PROD_ID, OLD_QTY, NEW_QTY, MOVED_BY, MOVE_DATE
  ) VALUES(:PRODUCT_ID, v_old, :QUANTITY, :APP_USER, SYSDATE);

  COMMIT;
  APEX_AI.SET_TOOL_RESULT(
    'Updated: ' || v_old || ' → ' || :QUANTITY || ' units'
  );
END;
```

Scenario: Upgrade**Q177** EXPERIENCED

SCENARIO: UPGRADE

You are deploying an APEX 24.2 app to APEX 26.1. What upgrade risks do you flag?**A N S W E R**

1. JS breaking change: regionStaticId → regionDomId, staticId → domId. Run Oracle diagnostic SQL first.
2. ORDS version: APEX 26.1 requires ORDS 26.1.1+.
3. Oracle DB version: APEX 26.1 requires Oracle 19c minimum. BOOLEAN type requires 26ai DB.
4. Universal Theme 26.1 template changes may affect custom CSS.
5. Custom plug-ins must be checked for compatibility.

E X A M P L E

```
-- Run BEFORE upgrading:
SELECT page_id, component_name, attribute_01
FROM apex_application_page_da_acts
WHERE application_id = :APP_ID
AND (attribute_01 LIKE '%regionStaticId%'
OR attribute_01 LIKE '%this.staticId%');

-- Fix: replace with regionDomId / domId

-- Verify ORDS version + DB version before upgrade
```

Scenario: Debug**Q178** EXPERIENCED

SCENARIO: DEBUG

A report using CONNECT BY runs fine in SQL Developer but breaks in APEX. Why?**A N S W E R**

APEX adds its own WHERE clause and pagination (ROWNUM) to report queries — this conflicts with CONNECT BY which has its own ordering (ORDER SIBLINGS BY).

Fix: Wrap the hierarchical query in a subquery, then apply APEX pagination on the outer query.

E X A M P L E

```
-- WRONG: CONNECT BY directly as IR source (APEX modifies it):
SELECT EMPNO, ENAME FROM EMP
START WITH MGR IS NULL
CONNECT BY PRIOR EMPNO = MGR;

-- CORRECT: wrap in subquery:
SELECT EMPNO, ENAME, MGR, ORG_LEVEL, ORG_CHART
FROM (
```

```
SELECT EMPNO, ENAME, MGR, LEVEL AS ORG_LEVEL,  
       LPAD(' ',2*(LEVEL-1)) || ENAME AS ORG_CHART  
FROM EMP  
START WITH MGR IS NULL  
CONNECT BY PRIOR EMPNO = MGR  
ORDER SIBLINGS BY ENAME  
)
```

Scenario: Architecture

Q179**EXPERIENCED****SCENARIO: ARCHITECTURE**

A user's session expires mid-wizard and they lose all wizard data. How do you prevent this?

ANSWER

1. Auto-save to staging table: After each step, MERGE data into WIZARD_STAGING. On session resume, check for saved data and reload.
2. Increase APEX Session Timeout (Shared Components → Security).
3. Heartbeat JavaScript: ping server every 10 minutes to keep session alive.
4. Warn user 5 minutes before timeout using a client-side timer DA.

EXAMPLE

```
-- Auto-save to staging (survives session expiry):  
MERGE INTO WIZARD_STAGING WS  
USING DUAL ON (WS.USER_ID = :APP_USER AND WS.STEP = :P_STEP)  
WHEN MATCHED THEN  
  UPDATE SET STEP_DATA = :P_STEP_JSON, UPDATED_ON = SYSDATE  
WHEN NOT MATCHED THEN  
  INSERT (USER_ID, STEP, STEP_DATA, CREATED_ON)  
  VALUES (:APP_USER, :P_STEP, :P_STEP_JSON, SYSDATE);  
  
-- Reload on return: check WIZARD_STAGING in Before Header process
```

Scenario: Performance

Q180**EXPERIENCED****SCENARIO: PERFORMANCE**

APEX app is used by 500 concurrent users and queries are timing out. What is your plan?

ANSWER

1. ORDS Connection Pool: Size correctly for 500 users in ords.xml.
2. EXPLAIN PLAN on all slow reports → add composite indexes.
3. Avoid N+1: pre-join lookup tables in report SQL instead of per-row queries.
4. Result Cache: Add RESULT_CACHE hint to static reference data.
5. Pagination: Verify all reports paginate — no report fetches 100K rows.
6. Materialized Views: Pre-compute summary dashboard data.

E X A M P L E

```
-- N+1 pattern to avoid (500 rows × 1 LOV query = 500 DB calls):  
-- BAD: LOV column in report queries DB once per row
```

```
-- GOOD: JOIN the lookup in the report SQL:  
SELECT O.ORDER_ID, C.CUST_NAME, S.STATUS_LABEL  
FROM ORDERS O  
JOIN CUSTOMERS C ON O.CUST_ID = C.CUST_ID  
JOIN ORDER_STATUSES S ON O.STATUS = S.STATUS_CODE  
-- 1 DB call instead of 500+
```

Scenario: Architecture**Q181****EXPERIENCED****SCENARIO: ARCHITECTURE****How do you build a dashboard that auto-refreshes KPI cards every 30 seconds?****A N S W E R**

Option 1 (Declarative): Region → Advanced → Auto Refresh = 30 seconds. Simplest.

Option 2 (JavaScript): DA on Page Load → Execute JavaScript → setInterval() to call apex.region().refresh() every 30 seconds.

E X A M P L E

```
-- Option 1: Declarative (simplest, zero code):  
-- KPI Region → Advanced → Auto Refresh → Enable = Yes  
-- Interval = 30 (seconds)
```

```
-- Option 2: JavaScript (more control over multiple regions):  
var interval = setInterval(function() {  
    apex.region('kpi_cards').refresh();  
    apex.region('revenue_chart').refresh();  
}, 30000);
```

```
// Cleanup on page unload:  
$(window).on('unload', function() {  
    clearInterval(interval);  
});
```

Q182**EXPERIENCED****SCENARIO: ARCHITECTURE**

How do you implement SSO (Single Sign-On) in an APEX application?

ANSWER

Options:

1. Oracle IDCS: Create App Registration in IDCS, configure APEX auth with OAuth2.
2. LDAP/Active Directory: Configure APEX LDAP authentication scheme.
3. SAML 2.0: Configure ORDS as a SAML Service Provider.
4. HTTP Header Variable: Enterprise proxy sets X-REMOTE-USER header after SSO.

EXAMPLE

-- LDAP/Active Directory setup:

Host: ldap.company.com

Port: 636 (SSL)

DN: cn=%LDAP_USER%,ou=employees,dc=company,dc=com

-- Post-authentication: set roles from LDAP:

-- Custom post-auth PL/SQL process:

BEGIN

-- Look up AD groups for current user

-- Set G_USER_ROLE Application Item

SET_APP_ITEM_FROM_AD_GROUPS(:APP_USER);

END;

Q183**EXPERIENCED****SCENARIO: ARCHITECTURE**

An APEX application needs to generate a PDF invoice on button click. Discuss architecture options.

ANSWER

Option 1 — APEX Document Generator (24.1+): Built-in. Create a Word template (.docx) with APEX tags. Use APEX_DATA_EXPORT. No external server.

Option 2 — AOP (APEX Office Print): Popular enterprise option. Word/Excel template + JSON data → AOP generates PDF. Requires AOP server.

Option 3 — Browser Print: Generate formatted HTML invoice → trigger window.print().

Option 4 — External PDF API: Call a PDF generation REST API from PL/SQL.

EXAMPLE

-- Option 3: Browser Print (simplest, no dependencies)

@media print {

.no-print { display: none; }

.invoice-body { width:100%; font-size:11pt; }

@page { margin:0.5in; size:A4; }

}

-- Open invoice page in new window:

apex.navigation.openInNewWindow(

'f?p=&APP_ID.:7:&SESSION.:NO::P7_SALE_ID:' + saleId

);

// User uses browser print dialog → saves as PDF

APEX AI

Q184 EXPERIENCED

APEX AI

How do you build an ad-hoc AI Agent using only PL/SQL via APEX_AI.GENERATE with p_tools?

ANSWER

APEX 26.1 introduced programmatic AI Agents via APEX_AI.GENERATE with p_tools parameter. Define tools inline in PL/SQL — no shared component needed. Each tool has a name, description, parameters, and a callback procedure.

EXAMPLE

```
DECLARE
  l_tools apex_ai.t_tools;
  l_response CLOB;
BEGIN
  l_tools := apex_ai.t_tools(
    apex_ai.t_tool(
      p_name      => 'GET_STOCK',
      p_description => 'Get current stock for a product',
      p_callback_procedure => 'pkg_ai.get_stock'
    )
  );
  l_response := APEX_AI.GENERATE(
    p_prompt      => :P3_USER_QUERY,
    p_service_static_id => 'router_cred',
    p_tools       => l_tools
  );
  :P3_AI_RESPONSE := l_response;
END;
```

APEX 26.1 Features

Q185 EXPERIENCED

APEX 26.1 FEATURES

What is the BOOLEAN session state data type in APEX 26.1? What database is required?

ANSWER

APEX 26.1 introduced native BOOLEAN data type for page items and application items. Previously boolean values were stored as VARCHAR2 ('Y'/'N'). Now you can declare items as BOOLEAN type and use them directly in PL/SQL.

Requirement: Oracle AI Database 26ai. NOT available on Oracle Database 19c.

EXAMPLE

```
-- Before APEX 26.1 (VARCHAR2 workaround):
IF :P3_IS_ACTIVE = 'Y' THEN
  -- do something
END IF;

-- APEX 26.1 + Oracle AI Database 26ai (native BOOLEAN):
IF :P3_IS_ACTIVE THEN -- direct boolean check
  -- do something
END IF;

-- Requires: Oracle AI Database 26ai
-- NOT available on Oracle Database 19c
```

APEX AI

Q186 EXPERIENCED

APEX AI

What is AI Vector Search in APEX and how do you implement it?

ANSWER

AI Vector Search finds semantically similar content using vector embeddings. Steps:

1. Add VECTOR column to your table.
2. Generate vector embeddings for existing data.
3. Create a Search Configuration in APEX Shared Components: Type = Oracle Vector Search.
4. Add a Search Results region — users type natural language, APEX finds similar records.

EXAMPLE

```
-- Vector similarity search SQL:
SELECT PRODUCT_NAME, DESCRIPTION, PRICE
FROM PRODUCTS
ORDER BY
  VECTOR_DISTANCE(
    DESCRIPTION_V,
    VECTOR_EMBEDDING(DOC_MODEL USING :P1_SEARCH AS DATA)
  )
FETCH FIRST 5 ROWS ONLY;
-- Returns 5 most semantically similar products
```


Q187 EXPERIENCED

APEX AI

What is the difference between an AI Configuration and a Generative AI Service?**A N S W E R**

Generative AI Service: Connection config — stores provider URL, model, and credential. Think of it as the 'connection string' to the AI.

AI Configuration: Application-level orchestration combining a Generative AI Service + System Prompt + RAG Sources + Tools. Defines HOW the AI behaves for a specific use case.

E X A M P L E

-- Generative AI Service (connection only):

Static ID: router_cred

Provider: OpenAI-Compatible

Model: google/gemini-2.5-flash-lite

-- AI Configuration (behaviour + connection):

Static ID: STOCK_MANAGER

Service: router_cred ← references service above

System: 'You are a POS inventory assistant...'

Tools: CHECK_STOCK, UPDATE_STOCK

-- Reference in AI Agent Dynamic Action

APEX 26.1 Features**Q188** EXPERIENCED

APEX 26.1 FEATURES

What is Describe Tables in APEX 26.1 and how does it improve LLM-based integrations?**A N S W E R**

Describe Tables generates human-readable, structured descriptions of database tables and views — including column names, data types, comments, and relationships. Formatted LLM-friendly. Used in NL2IR so the AI understands your specific schema. Significantly improves NL2SQL accuracy — AI gets schema context, not just raw column names.

E X A M P L E

-- Describe Tables output:

Table: POS_INVENTORY

Description: Tracks real-time stock for all products

Columns:

PROD_ID (NUMBER) - Unique product identifier

PROD_NAME (VARCHAR2) - Product display name

STOCK_QTY (NUMBER) - Current quantity in stock

CATEGORY_ID (NUMBER) - FK to POS_CATEGORIES.CAT_ID

-- AI now understands schema context for NL2SQL

Performance Tuning

Q189**EXPERIENCED**

PERFORMANCE TUNING

What is the difference between a Materialized View and a View for APEX reporting?

ANSWER

View: Stored SQL query. Executes the full underlying SQL every time it is queried. Always returns current data. Zero storage.

Materialized View (MV): Pre-computes and STORES the result as a physical table. Querying it is as fast as querying a table. Refreshed on schedule or on demand. Use MVs for expensive aggregation reports where real-time is not critical.

EXAMPLE

-- Regular View (always current, can be slow):

```
CREATE VIEW V_DAILY_SALES AS
SELECT TRUNC(SALE_DATE) D, SUM(TOTAL) REVENUE
FROM SALES GROUP BY TRUNC(SALE_DATE);
```

-- Materialized View (fast, refreshes nightly):

```
CREATE MATERIALIZED VIEW MV_DAILY_SALES
REFRESH COMPLETE ON DEMAND AS
SELECT TRUNC(SALE_DATE) AS DAY, SUM(TOTAL) AS REVENUE
FROM SALES GROUP BY TRUNC(SALE_DATE);
```

-- Query runs instantly — reads pre-computed table

Scenario: Architecture

Q190**EXPERIENCED**

SCENARIO: ARCHITECTURE

How do you prevent unauthorized direct URL access to APEX pages?

ANSWER

1. Authorization Scheme on the page — users without the required role see an error page.
2. Authentication — ensure the app requires login.
3. Modal Dialog pages — cannot be accessed directly via URL.
4. Session State Protection — prevents URL parameter tampering.
5. Server-side data checks in processes — defence in depth.

EXAMPLE

-- Page level:

-- Page → Security → Authorization Scheme = IS_MANAGER

-- Region level (hide sensitive data within a page):

-- Region → Security → Authorization Scheme = IS_ADMIN

```
-- Defence in depth — check in the process too:  
IF NOT APEX_UTIL.IS_IN_GROUP('MANAGER') THEN  
  RAISE_APPLICATION_ERROR(-20001, 'Access denied');  
END IF;
```

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